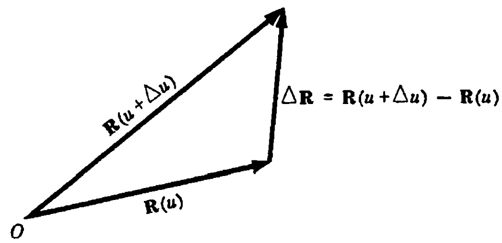
Chapter 3 VECTOR DIFFERENTIATION

ORDINARY DERIVATIVES OF VECTORS. Let $\mathbf{R}(u)$ be a vector depending on a single scalar variable u. Then

$$\frac{\Delta \mathbf{R}}{\Delta u} = \frac{\mathbf{R}(u+\Delta u) - \mathbf{R}(u)}{\Delta u}$$

where Δu denotes an increment in u (see adjoining figure).



The ordinary derivative of the vector $\mathbf{R}(u)$ with respect to the scalar u is given by

$$\frac{d\mathbf{R}}{du} = \lim_{\Delta u \to 0} \frac{\Delta \mathbf{R}}{\Delta u} = \lim_{\Delta u \to 0} \frac{\mathbf{R}(u+\Delta u) - \mathbf{R}(u)}{\Delta u}$$

if the limit exists.

Since $\dfrac{d\mathbf{R}}{du}$ is itself a vector depending on u, we can consider its derivative with respect to u. If this derivative exists it is denoted by $\dfrac{d^2\mathbf{R}}{du^2}$. In like manner higher order derivatives are described.

SPACE CURVES. If in particular $\mathbf{R}(u)$ is the position vector $\mathbf{r}(u)$ joining the origin O of a coordinate system and any point (x,y,z), then

$$\mathbf{r}(u) = x(u)\mathbf{i} + y(u)\mathbf{j} + z(u)\mathbf{k}$$

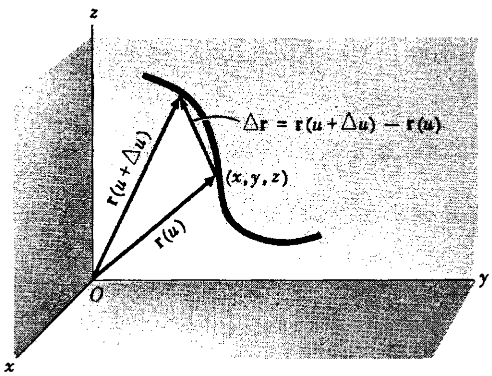
and specification of the vector function $\mathbf{r}(u)$ defines x, y and z as functions of u.

As u changes, the terminal point of $\mathbf{r}$ describes a *space curve* having parametric equations

$$x = x(u), \quad y = y(u), \quad z = z(u)$$

Then $\dfrac{\Delta \mathbf{r}}{\Delta u} = \dfrac{\mathbf{r}(u+\Delta u) - \mathbf{r}(u)}{\Delta u}$ is a vector in the direction of $\Delta \mathbf{r}$ (see adjacent figure). If $\displaystyle\lim_{\Delta u \to 0} \frac{\Delta \mathbf{r}}{\Delta u} = \frac{d\mathbf{r}}{du}$ exists, the limit will be a vector in the direction of the tangent to the space curve at (x,y,z) and is given by

$$\frac{d\mathbf{r}}{du} = \frac{dx}{du}\mathbf{i} + \frac{dy}{du}\mathbf{j} + \frac{dz}{du}\mathbf{k}$$



If u is the time t, $\dfrac{d\mathbf{r}}{dt}$ represents the *velocity* $\mathbf{v}$ with which the terminal point of $\mathbf{r}$ describes the curve. Similarly, $\dfrac{d\mathbf{v}}{dt} = \dfrac{d^2\mathbf{r}}{dt^2}$ represents its *acceleration* $\mathbf{a}$ along the curve.

CONTINUITY AND DIFFERENTIABILITY. A scalar function $\phi(u)$ is called *continuous* at u if $\lim_{\Delta u \to 0} \phi(u+\Delta u) = \phi(u)$. Equivalently, $\phi(u)$ is continuous at u if for each positive number ϵ we can find some positive number δ such that

$$|\phi(u+\Delta u) - \phi(u)| < \epsilon \quad \text{whenever} \quad |\Delta u| < \delta.$$

A vector function $\mathbf{R}(u) = R_1(u)\mathbf{i} + R_2(u)\mathbf{j} + R_3(u)\mathbf{k}$ is called *continuous* at u if the three scalar functions $R_1(u)$, $R_2(u)$ and $R_3(u)$ are continuous at u or if $\lim_{\Delta u \to 0} \mathbf{R}(u+\Delta u) = \mathbf{R}(u)$. Equivalently, $\mathbf{R}(u)$ is continuous at u if for each positive number ϵ we can find some positive number δ such that

$$|\mathbf{R}(u+\Delta u) - \mathbf{R}(u)| < \epsilon \quad \text{whenever} \quad |\Delta u| < \delta.$$

A scalar or vector function of u is called *differentiable of order n* if its nth derivative exists. A function which is differentiable is necessarily continuous but the converse is not true. Unless otherwise stated we assume that all functions considered are differentiable to any order needed in a particular discussion.

DIFFERENTIATION FORMULAS. If $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are differentiable vector functions of a scalar u, and ϕ is a differentiable scalar function of u, then

1. $\dfrac{d}{du}(\mathbf{A}+\mathbf{B}) = \dfrac{d\mathbf{A}}{du} + \dfrac{d\mathbf{B}}{du}$

2. $\dfrac{d}{du}(\mathbf{A}\cdot\mathbf{B}) = \mathbf{A}\cdot\dfrac{d\mathbf{B}}{du} + \dfrac{d\mathbf{A}}{du}\cdot\mathbf{B}$

3. $\dfrac{d}{du}(\mathbf{A}\times\mathbf{B}) = \mathbf{A}\times\dfrac{d\mathbf{B}}{du} + \dfrac{d\mathbf{A}}{du}\times\mathbf{B}$

4. $\dfrac{d}{du}(\phi\mathbf{A}) = \phi\dfrac{d\mathbf{A}}{du} + \dfrac{d\phi}{du}\mathbf{A}$

5. $\dfrac{d}{du}(\mathbf{A}\cdot\mathbf{B}\times\mathbf{C}) = \mathbf{A}\cdot\mathbf{B}\times\dfrac{d\mathbf{C}}{du} + \mathbf{A}\cdot\dfrac{d\mathbf{B}}{du}\times\mathbf{C} + \dfrac{d\mathbf{A}}{du}\cdot\mathbf{B}\times\mathbf{C}$

6. $\dfrac{d}{du}\{\mathbf{A}\times(\mathbf{B}\times\mathbf{C})\} = \mathbf{A}\times(\mathbf{B}\times\dfrac{d\mathbf{C}}{du}) + \mathbf{A}\times(\dfrac{d\mathbf{B}}{du}\times\mathbf{C}) + \dfrac{d\mathbf{A}}{du}\times(\mathbf{B}\times\mathbf{C})$

The order in these products may be important.

PARTIAL DERIVATIVES OF VECTORS. If $\mathbf{A}$ is a vector depending on more than one scalar variable, say x, y, z for example, then we write $\mathbf{A} = \mathbf{A}(x,y,z)$. The partial derivative of $\mathbf{A}$ with respect to x is defined as

$$\frac{\partial \mathbf{A}}{\partial x} = \lim_{\Delta x \to 0} \frac{\mathbf{A}(x+\Delta x, y, z) - \mathbf{A}(x,y,z)}{\Delta x}$$

if this limit exists. Similarly,

$$\frac{\partial \mathbf{A}}{\partial y} = \lim_{\Delta y \to 0} \frac{\mathbf{A}(x, y+\Delta y, z) - \mathbf{A}(x,y,z)}{\Delta y}$$

$$\frac{\partial \mathbf{A}}{\partial z} = \lim_{\Delta z \to 0} \frac{\mathbf{A}(x,y, z+\Delta z) - \mathbf{A}(x,y,z)}{\Delta z}$$

are the partial derivatives of **A** with respect to y and z respectively if these limits exist.

The remarks on continuity and differentiability for functions of one variable can be extended to functions of two or more variables. For example, $\phi(x,y)$ is called continuous at (x,y) if $\lim\limits_{\substack{\Delta x \to 0 \\ \Delta y \to 0}} \phi(x+\Delta x,\, y+\Delta y) = \phi(x,y)$, or if for each positive number ϵ we can find some positive number δ such that $|\phi(x+\Delta x,\, y+\Delta y) - \phi(x,y)| < \epsilon$ whenever $|\Delta x| < \delta$ and $|\Delta y| < \delta$. Similar definitions hold for vector functions.

For functions of two or more variables we use the term *differentiable* to mean that the function has continuous first partial derivatives. (The term is used by others in a slightly weaker sense.)

Higher derivatives can be defined as in the calculus. Thus, for example,

$$\frac{\partial^2 \mathbf{A}}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial \mathbf{A}}{\partial x}\right), \quad \frac{\partial^2 \mathbf{A}}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial \mathbf{A}}{\partial y}\right), \quad \frac{\partial^2 \mathbf{A}}{\partial z^2} = \frac{\partial}{\partial z}\left(\frac{\partial \mathbf{A}}{\partial z}\right)$$

$$\frac{\partial^2 \mathbf{A}}{\partial x\, \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial \mathbf{A}}{\partial y}\right), \quad \frac{\partial^2 \mathbf{A}}{\partial y\, \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial \mathbf{A}}{\partial x}\right), \quad \frac{\partial^3 \mathbf{A}}{\partial x\, \partial z^2} = \frac{\partial}{\partial x}\left(\frac{\partial^2 \mathbf{A}}{\partial z^2}\right)$$

If **A** has continuous partial derivatives of the second order at least, then $\dfrac{\partial^2 \mathbf{A}}{\partial x\, \partial y} = \dfrac{\partial^2 \mathbf{A}}{\partial y\, \partial x}$, i.e. the order of differentiation does not matter.

Rules for partial differentiation of vectors are similar to those used in elementary calculus for scalar functions. Thus if **A** and **B** are functions of x,y,z then, for example,

1. $\dfrac{\partial}{\partial x}(\mathbf{A}\cdot\mathbf{B}) = \mathbf{A}\cdot\dfrac{\partial \mathbf{B}}{\partial x} + \dfrac{\partial \mathbf{A}}{\partial x}\cdot\mathbf{B}$

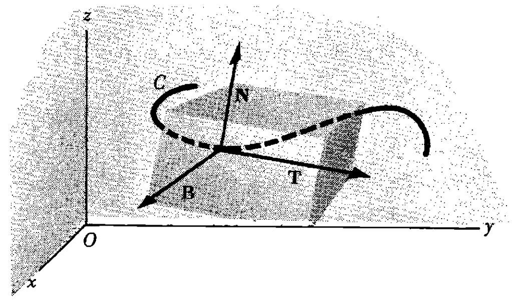
2. $\dfrac{\partial}{\partial x}(\mathbf{A}\times\mathbf{B}) = \mathbf{A}\times\dfrac{\partial \mathbf{B}}{\partial x} + \dfrac{\partial \mathbf{A}}{\partial x}\times\mathbf{B}$

3. $$\frac{\partial^2}{\partial y\, \partial x}(\mathbf{A}\cdot\mathbf{B}) = \frac{\partial}{\partial y}\left\{\frac{\partial}{\partial x}(\mathbf{A}\cdot\mathbf{B})\right\} = \frac{\partial}{\partial y}\left\{\mathbf{A}\cdot\frac{\partial \mathbf{B}}{\partial x} + \frac{\partial \mathbf{A}}{\partial x}\cdot\mathbf{B}\right\}$$
$$= \mathbf{A}\cdot\frac{\partial^2 \mathbf{B}}{\partial y\, \partial x} + \frac{\partial \mathbf{A}}{\partial y}\cdot\frac{\partial \mathbf{B}}{\partial x} + \frac{\partial \mathbf{A}}{\partial x}\cdot\frac{\partial \mathbf{B}}{\partial y} + \frac{\partial^2 \mathbf{A}}{\partial y\, \partial x}\cdot\mathbf{B}, \quad \text{etc.}$$

DIFFERENTIALS OF VECTORS follow rules similar to those of elementary calculus. For example,

1. If $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$, then $d\mathbf{A} = dA_1\mathbf{i} + dA_2\mathbf{j} + dA_3\mathbf{k}$
2. $d(\mathbf{A}\cdot\mathbf{B}) = \mathbf{A}\cdot d\mathbf{B} + d\mathbf{A}\cdot\mathbf{B}$
3. $d(\mathbf{A}\times\mathbf{B}) = \mathbf{A}\times d\mathbf{B} + d\mathbf{A}\times\mathbf{B}$
4. If $\mathbf{A} = \mathbf{A}(x,y,z)$, then $d\mathbf{A} = \dfrac{\partial \mathbf{A}}{\partial x}dx + \dfrac{\partial \mathbf{A}}{\partial y}dy + \dfrac{\partial \mathbf{A}}{\partial z}dz$, etc.

DIFFERENTIAL GEOMETRY involves a study of space curves and surfaces. If C is a space curve defined by the function $\mathbf{r}(u)$, then we have seen that $\dfrac{d\mathbf{r}}{du}$ is a vector in the direction of the tangent to C. If the scalar u is taken as the arc length s measured from some fixed point on C, then $\dfrac{d\mathbf{r}}{ds}$ is a unit tangent vector to C and is denoted by $\mathbf{T}$ (see diagram below). The rate at which $\mathbf{T}$ changes with respect to s is a measure of the curvature of C and is given by $\frac{d\mathbf{T}}{ds}$. The direction of $\frac{d\mathbf{T}}{ds}$ at any given point on C is normal to the curve at that point (see Problem 9). If $\mathbf{N}$ is a unit vector in this normal direction, it is called the *principal normal* to the curve. Then $\frac{d\mathbf{T}}{ds} = \kappa\mathbf{N}$, where κ is called the *curvature* of C at the specified point. The quantity $\rho = 1/\kappa$ is called the *radius of curvature*.



A unit vector $\mathbf{B}$ perpendicular to the plane of $\mathbf{T}$ and $\mathbf{N}$ and such that $\mathbf{B} = \mathbf{T}\times\mathbf{N}$, is called the *binormal* to the curve. It follows that directions $\mathbf{T}, \mathbf{N}, \mathbf{B}$ form a localized right-handed rectangular coordinate system at any specified point of C. This coordinate system is called the *trihedral* or *triad* at the point. As s changes, the coordinate system moves and is known as the *moving trihedral.*

A set of relations involving derivatives of the fundamental vectors $\mathbf{T}, \mathbf{N}$ and $\mathbf{B}$ is known collectively as the *Frenet-Serret formulas* given by

$$\frac{d\mathbf{T}}{ds} = \kappa\mathbf{N}, \qquad \frac{d\mathbf{N}}{ds} = \tau\mathbf{B} - \kappa\mathbf{T}, \qquad \frac{d\mathbf{B}}{ds} = -\tau\mathbf{N}$$

where τ is a scalar called the *torsion*. The quantity $\sigma = 1/\tau$ is called the *radius of torsion.*

The *osculating plane* to a curve at a point P is the plane containing the tangent and principal normal at P. The *normal plane* is the plane through P perpendicular to the tangent. The *rectifying plane* is the plane through P which is perpendicular to the principal normal.

MECHANICS often includes a study of the motion of particles along curves, this study being known as *kinematics*. In this connection some of the results of differential geometry can be of value.

A study of forces on moving objects is considered in *dynamics*. Fundamental to this study is Newton's famous law which states that if $\mathbf{F}$ is the net force acting on an object of mass m moving with velocity $\mathbf{v}$, then

$$\mathbf{F} = \frac{d}{dt}(m\mathbf{v})$$

where $m\mathbf{v}$ is the momentum of the object. If m is constant this becomes $\mathbf{F} = m\frac{d\mathbf{v}}{dt} = m\mathbf{a}$, where $\mathbf{a}$ is the acceleration of the object.

SOLVED PROBLEMS

1. If $\mathbf{R}(u) = x(u)\mathbf{i} + y(u)\mathbf{j} + z(u)\mathbf{k}$, where x, y and z are differentiable functions of a scalar u, prove that $\dfrac{d\mathbf{R}}{du} = \dfrac{dx}{du}\mathbf{i} + \dfrac{dy}{du}\mathbf{j} + \dfrac{dz}{du}\mathbf{k}$.

$$\begin{aligned}\frac{d\mathbf{R}}{du} &= \lim_{\Delta u\to 0} \frac{\mathbf{R}(u+\Delta u) - \mathbf{R}(u)}{\Delta u}\\ &= \lim_{\Delta u\to 0} \frac{[x(u+\Delta u)\mathbf{i} + y(u+\Delta u)\mathbf{j} + z(u+\Delta u)\mathbf{k}] - [x(u)\mathbf{i} + y(u)\mathbf{j} + z(u)\mathbf{k}]}{\Delta u}\\ &= \lim_{\Delta u\to 0} \frac{x(u+\Delta u) - x(u)}{\Delta u}\mathbf{i} + \frac{y(u+\Delta u) - y(u)}{\Delta u}\mathbf{j} + \frac{z(u+\Delta u) - z(u)}{\Delta u}\mathbf{k}\\ &= \frac{dx}{du}\mathbf{i} + \frac{dy}{du}\mathbf{j} + \frac{dz}{du}\mathbf{k}\end{aligned}$$

2. Given $\mathbf{R} = \sin t\,\mathbf{i} + \cos t\,\mathbf{j} + t\mathbf{k}$, find (a) $\dfrac{d\mathbf{R}}{dt}$, (b) $\dfrac{d^2\mathbf{R}}{dt^2}$, (c) $\left|\dfrac{d\mathbf{R}}{dt}\right|$, (d) $\left|\dfrac{d^2\mathbf{R}}{dt^2}\right|$.

(a) $\dfrac{d\mathbf{R}}{dt} = \dfrac{d}{dt}(\sin t)\mathbf{i} + \dfrac{d}{dt}(\cos t)\mathbf{j} + \dfrac{d}{dt}(t)\mathbf{k} = \cos t\,\mathbf{i} - \sin t\,\mathbf{j} + \mathbf{k}$

(b) $\dfrac{d^2\mathbf{R}}{dt^2} = \dfrac{d}{dt}\left(\dfrac{d\mathbf{R}}{dt}\right) = \dfrac{d}{dt}(\cos t)\mathbf{i} - \dfrac{d}{dt}(\sin t)\mathbf{j} + \dfrac{d}{dt}(1)\mathbf{k} = -\sin t\,\mathbf{i} - \cos t\,\mathbf{j}$

(c) $\left|\dfrac{d\mathbf{R}}{dt}\right| = \sqrt{(\cos t)^2 + (-\sin t)^2 + (1)^2} = \sqrt{2}$

(d) $\left|\dfrac{d^2\mathbf{R}}{dt^2}\right| = \sqrt{(-\sin t)^2 + (-\cos t)^2} = 1$

3. A particle moves along a curve whose parametric equations are $x = e^{-t}$, $y = 2\cos 3t$, $z = 2\sin 3t$, where t is the time.
(a) Determine its velocity and acceleration at any time.
(b) Find the magnitudes of the velocity and acceleration at $t = 0$.

(a) The position vector $\mathbf{r}$ of the particle is $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = e^{-t}\mathbf{i} + 2\cos 3t\,\mathbf{j} + 2\sin 3t\,\mathbf{k}$.

Then the velocity is $\mathbf{v} = \dfrac{d\mathbf{r}}{dt} = -e^{-t}\mathbf{i} - 6\sin 3t\,\mathbf{j} + 6\cos 3t\,\mathbf{k}$

and the acceleration is $\mathbf{a} = \dfrac{d^2\mathbf{r}}{dt^2} = e^{-t}\mathbf{i} - 18\cos 3t\,\mathbf{j} - 18\sin 3t\,\mathbf{k}$

(b) At $t = 0$, $\dfrac{d\mathbf{r}}{dt} = -\mathbf{i} + 6\mathbf{k}$ and $\dfrac{d^2\mathbf{r}}{dt^2} = \mathbf{i} - 18\mathbf{j}$. Then

magnitude of velocity at $t = 0$ is $\sqrt{(-1)^2 + (6)^2} = \sqrt{37}$

magnitude of acceleration at $t = 0$ is $\sqrt{(1)^2 + (-18)^2} = \sqrt{325}$.

4. A particle moves along the curve $x = 2t^2$, $y = t^2 - 4t$, $z = 3t - 5$, where t is the time. Find the components of its velocity and acceleration at time $t = 1$ in the direction $\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$.

$$\text{Velocity} = \frac{d\mathbf{r}}{dt} = \frac{d}{dt}[2t^2\mathbf{i} + (t^2-4t)\mathbf{j} + (3t-5)\mathbf{k}]$$
$$= 4t\,\mathbf{i} + (2t-4)\mathbf{j} + 3\mathbf{k} = 4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k} \quad \text{at } t = 1.$$

Unit vector in direction $\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$ is $\dfrac{\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}}{\sqrt{(1)^2 + (-3)^2 + (2)^2}} = \dfrac{\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}}{\sqrt{14}}$.

Then the component of the velocity in the given direction is

$$\frac{(4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k})\cdot(\mathbf{i} - 3\mathbf{j} + 2\mathbf{k})}{\sqrt{14}} = \frac{(4)(1) + (-2)(-3) + (3)(2)}{\sqrt{14}} = \frac{16}{\sqrt{14}} = \frac{8\sqrt{14}}{7}$$

$$\text{Acceleration} = \frac{d^2\mathbf{r}}{dt^2} = \frac{d}{dt}\left(\frac{d\mathbf{r}}{dt}\right) = \frac{d}{dt}[4t\,\mathbf{i} + (2t-4)\mathbf{j} + 3\mathbf{k}] = 4\mathbf{i} + 2\mathbf{j} + 0\mathbf{k}.$$

Then the component of the acceleration in the given direction is

$$\frac{(4\mathbf{i} + 2\mathbf{j} + 0\mathbf{k})\cdot(\mathbf{i} - 3\mathbf{j} + 2\mathbf{k})}{\sqrt{14}} = \frac{(4)(1) + (2)(-3) + (0)(2)}{\sqrt{14}} = \frac{-2}{\sqrt{14}} = \frac{-\sqrt{14}}{7}$$

5. A curve C is defined by parametric equations $x = x(s)$, $y = y(s)$, $z = z(s)$, where s is the arc length of C measured from a fixed point on C. If $\mathbf{r}$ is the position vector of any point on C, show that $d\mathbf{r}/ds$ is a unit vector tangent to C.

The vector $\dfrac{d\mathbf{r}}{ds} = \dfrac{d}{ds}(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = \dfrac{dx}{ds}\mathbf{i} + \dfrac{dy}{ds}\mathbf{j} + \dfrac{dz}{ds}\mathbf{k}$ is tangent to the curve $x = x(s)$, $y = y(s)$, $z = z(s)$. To show that it has unit magnitude we note that

$$\left|\frac{d\mathbf{r}}{ds}\right| = \sqrt{\left(\frac{dx}{ds}\right)^2 + \left(\frac{dy}{ds}\right)^2 + \left(\frac{dz}{ds}\right)^2} = \sqrt{\frac{(dx)^2 + (dy)^2 + (dz)^2}{(ds)^2}} = 1$$

since $(ds)^2 = (dx)^2 + (dy)^2 + (dz)^2$ from the calculus.

6. (*a*) Find the unit tangent vector to any point on the curve $x = t^2+1$, $y = 4t-3$, $z = 2t^2-6t$.
(*b*) Determine the unit tangent at the point where $t = 2$.

(*a*) A tangent vector to the curve at any point is

$$\frac{d\mathbf{r}}{dt} = \frac{d}{dt}[(t^2+1)\mathbf{i} + (4t-3)\mathbf{j} + (2t^2-6t)\mathbf{k}] = 2t\,\mathbf{i} + 4\mathbf{j} + (4t-6)\mathbf{k}$$

The magnitude of the vector is $\left|\dfrac{d\mathbf{r}}{dt}\right| = \sqrt{(2t)^2 + (4)^2 + (4t-6)^2}$.

Then the required unit tangent vector is $\mathbf{T} = \dfrac{2t\,\mathbf{i} + 4\mathbf{j} + (4t-6)\mathbf{k}}{\sqrt{(2t)^2 + (4)^2 + (4t-6)^2}}$

Note that since $\left|\dfrac{d\mathbf{r}}{dt}\right| = \dfrac{ds}{dt}$, $\mathbf{T} = \dfrac{d\mathbf{r}/dt}{ds/dt} = \dfrac{d\mathbf{r}}{ds}$.

(*b*) At $t = 2$, the unit tangent vector is $\mathbf{T} = \dfrac{4\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}}{\sqrt{(4)^2 + (4)^2 + (2)^2}} = \dfrac{2}{3}\mathbf{i} + \dfrac{2}{3}\mathbf{j} + \dfrac{1}{3}\mathbf{k}$.

7. If $\mathbf{A}$ and $\mathbf{B}$ are differentiable functions of a scalar u, prove:

(*a*) $\dfrac{d}{du}(\mathbf{A}\cdot\mathbf{B}) = \mathbf{A}\cdot\dfrac{d\mathbf{B}}{du} + \dfrac{d\mathbf{A}}{du}\cdot\mathbf{B}$, (*b*) $\dfrac{d}{du}(\mathbf{A}\times\mathbf{B}) = \mathbf{A}\times\dfrac{d\mathbf{B}}{du} + \dfrac{d\mathbf{A}}{du}\times\mathbf{B}$

$$(a)\ \frac{d}{du}(\mathbf{A}\cdot\mathbf{B}) = \lim_{\Delta u\to 0} \frac{(\mathbf{A}+\Delta\mathbf{A})\cdot(\mathbf{B}+\Delta\mathbf{B}) - \mathbf{A}\cdot\mathbf{B}}{\Delta u}$$

$$= \lim_{\Delta u\to 0} \frac{\mathbf{A}\cdot\Delta\mathbf{B} + \Delta\mathbf{A}\cdot\mathbf{B} + \Delta\mathbf{A}\cdot\Delta\mathbf{B}}{\Delta u}$$

$$= \lim_{\Delta u\to 0} \mathbf{A}\cdot\frac{\Delta\mathbf{B}}{\Delta u} + \frac{\Delta\mathbf{A}}{\Delta u}\cdot\mathbf{B} + \frac{\Delta\mathbf{A}}{\Delta u}\cdot\Delta\mathbf{B} = \mathbf{A}\cdot\frac{d\mathbf{B}}{du} + \frac{d\mathbf{A}}{du}\cdot\mathbf{B}$$

Another Method. Let $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$, $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$. Then

$$\frac{d}{du}(\mathbf{A}\cdot\mathbf{B}) = \frac{d}{du}(A_1B_1 + A_2B_2 + A_3B_3)$$

$$= (A_1\frac{dB_1}{du} + A_2\frac{dB_2}{du} + A_3\frac{dB_3}{du}) + (\frac{dA_1}{du}B_1 + \frac{dA_2}{du}B_2 + \frac{dA_3}{du}B_3) = \mathbf{A}\cdot\frac{d\mathbf{B}}{du} + \frac{d\mathbf{A}}{du}\cdot\mathbf{B}$$

$$(b)\ \frac{d}{du}(\mathbf{A}\times\mathbf{B}) = \lim_{\Delta u\to 0} \frac{(\mathbf{A}+\Delta\mathbf{A})\times(\mathbf{B}+\Delta\mathbf{B}) - \mathbf{A}\times\mathbf{B}}{\Delta u}$$

$$= \lim_{\Delta u\to 0} \frac{\mathbf{A}\times\Delta\mathbf{B} + \Delta\mathbf{A}\times\mathbf{B} + \Delta\mathbf{A}\times\Delta\mathbf{B}}{\Delta u}$$

$$= \lim_{\Delta u\to 0} \mathbf{A}\times\frac{\Delta\mathbf{B}}{\Delta u} + \frac{\Delta\mathbf{A}}{\Delta u}\times\mathbf{B} + \frac{\Delta\mathbf{A}}{\Delta u}\times\Delta\mathbf{B} = \mathbf{A}\times\frac{d\mathbf{B}}{du} + \frac{d\mathbf{A}}{du}\times\mathbf{B}$$

Another Method.

$$\frac{d}{du}(\mathbf{A}\times\mathbf{B}) = \frac{d}{du}\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix}$$

Using a theorem on differentiation of a determinant, this becomes

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_1 & A_2 & A_3 \\ \frac{dB_1}{du} & \frac{dB_2}{du} & \frac{dB_3}{du} \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{dA_1}{du} & \frac{dA_2}{du} & \frac{dA_3}{du} \\ B_1 & B_2 & B_3 \end{vmatrix} = \mathbf{A}\times\frac{d\mathbf{B}}{du} + \frac{d\mathbf{A}}{du}\times\mathbf{B}$$

8. If $\mathbf{A} = 5t^2\mathbf{i} + t\mathbf{j} - t^3\mathbf{k}$ and $\mathbf{B} = \sin t\ \mathbf{i} - \cos t\ \mathbf{j}$, find (a) $\frac{d}{dt}(\mathbf{A}\cdot\mathbf{B})$, (b) $\frac{d}{dt}(\mathbf{A}\times\mathbf{B})$, (c) $\frac{d}{dt}(\mathbf{A}\cdot\mathbf{A})$.

$$(a)\ \frac{d}{dt}(\mathbf{A}\cdot\mathbf{B}) = \mathbf{A}\cdot\frac{d\mathbf{B}}{dt} + \frac{d\mathbf{A}}{dt}\cdot\mathbf{B}$$

$$= (5t^2\mathbf{i} + t\mathbf{j} - t^3\mathbf{k})\cdot(\cos t\ \mathbf{i} + \sin t\ \mathbf{j}) + (10t\,\mathbf{i} + \mathbf{j} - 3t^2\mathbf{k})\cdot(\sin t\ \mathbf{i} - \cos t\ \mathbf{j})$$

$$= 5t^2\cos t + t\sin t + 10t\sin t - \cos t = (5t^2-1)\cos t + 11t\sin t$$

Another Method. $\mathbf{A}\cdot\mathbf{B} = 5t^2\sin t - t\cos t$. Then

$$\frac{d}{dt}(\mathbf{A}\cdot\mathbf{B}) = \frac{d}{dt}(5t^2\sin t - t\cos t) = 5t^2\cos t + 10t\sin t + t\sin t - \cos t$$
$$= (5t^2-1)\cos t + 11t\sin t$$

$$(b)\ \frac{d}{dt}(\mathbf{A}\times\mathbf{B}) = \mathbf{A}\times\frac{d\mathbf{B}}{dt} + \frac{d\mathbf{A}}{dt}\times\mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 5t^2 & t & -t^3 \\ \cos t & \sin t & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 10t & 1 & -3t^2 \\ \sin t & -\cos t & 0 \end{vmatrix}$$

$$= [t^3 \sin t\,\mathbf{i} - t^3 \cos t\,\mathbf{j} + (5t^2 \sin t - t\cos t)\mathbf{k}] + [-3t^2 \cos t\,\mathbf{i} - 3t^2 \sin t\,\mathbf{j} + (-10t\cos t - \sin t)\mathbf{k}]$$

$$= (t^3 \sin t - 3t^2 \cos t)\mathbf{i} - (t^3 \cos t + 3t^2 \sin t)\mathbf{j} + (5t^2 \sin t - \sin t - 11t\cos t)\mathbf{k}$$

Another Method.

$$\mathbf{A}\times\mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 5t^2 & t & -t^3 \\ \sin t & -\cos t & 0 \end{vmatrix} = -t^3\cos t\,\mathbf{i} - t^3 \sin t\,\mathbf{j} + (-5t^2\cos t - t\sin t)\mathbf{k}$$

Then $\dfrac{d}{dt}(\mathbf{A}\times\mathbf{B}) = (t^3\sin t - 3t^2\cos t)\mathbf{i} - (t^3\cos t + 3t^2\sin t)\mathbf{j} + (5t^2\sin t - 11t\cos t - \sin t)\mathbf{k}$

(*c*) $\dfrac{d}{dt}(\mathbf{A}\cdot\mathbf{A}) = \mathbf{A}\cdot\dfrac{d\mathbf{A}}{dt} + \dfrac{d\mathbf{A}}{dt}\cdot\mathbf{A} = 2\mathbf{A}\cdot\dfrac{d\mathbf{A}}{dt}$

$$= 2(5t^2\mathbf{i} + t\mathbf{j} - t^3\mathbf{k})\cdot(10t\,\mathbf{i} + \mathbf{j} - 3t^2\mathbf{k}) = 100t^3 + 2t + 6t^5$$

Another Method. $\mathbf{A}\cdot\mathbf{A} = (5t^2)^2 + (t)^2 + (-t^3)^2 = 25t^4 + t^2 + t^6$

Then $\dfrac{d}{dt}(25t^4 + t^2 + t^6) = 100t^3 + 2t + 6t^5$.

9. If $\mathbf{A}$ has constant magnitude show that $\mathbf{A}$ and $d\mathbf{A}/dt$ are perpendicular provided $|d\mathbf{A}/dt| \neq 0$.

Since $\mathbf{A}$ has constant magnitude, $\mathbf{A}\cdot\mathbf{A}$ = constant.

Then $\dfrac{d}{dt}(\mathbf{A}\cdot\mathbf{A}) = \mathbf{A}\cdot\dfrac{d\mathbf{A}}{dt} + \dfrac{d\mathbf{A}}{dt}\cdot\mathbf{A} = 2\mathbf{A}\cdot\dfrac{d\mathbf{A}}{dt} = 0$.

Thus $\mathbf{A}\cdot\dfrac{d\mathbf{A}}{dt} = 0$ and $\mathbf{A}$ is perpendicular to $\dfrac{d\mathbf{A}}{dt}$ provided $\left|\dfrac{d\mathbf{A}}{dt}\right| \neq 0$.

10. Prove $\dfrac{d}{du}(\mathbf{A}\cdot\mathbf{B}\times\mathbf{C}) = \mathbf{A}\cdot\mathbf{B}\times\dfrac{d\mathbf{C}}{du} + \mathbf{A}\cdot\dfrac{d\mathbf{B}}{du}\times\mathbf{C} + \dfrac{d\mathbf{A}}{du}\cdot\mathbf{B}\times\mathbf{C}$, where $\mathbf{A},\mathbf{B},\mathbf{C}$ are differentiable functions of a scalar u.

By Problems 7(*a*) and 7(*b*), $\dfrac{d}{du}\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \mathbf{A}\cdot\dfrac{d}{du}(\mathbf{B}\times\mathbf{C}) + \dfrac{d\mathbf{A}}{du}\cdot\mathbf{B}\times\mathbf{C}$

$$= \mathbf{A}\cdot[\mathbf{B}\times\frac{d\mathbf{C}}{du} + \frac{d\mathbf{B}}{du}\times\mathbf{C}] + \frac{d\mathbf{A}}{du}\cdot\mathbf{B}\times\mathbf{C}$$

$$= \mathbf{A}\cdot\mathbf{B}\times\frac{d\mathbf{C}}{du} + \mathbf{A}\cdot\frac{d\mathbf{B}}{du}\times\mathbf{C} + \frac{d\mathbf{A}}{du}\cdot\mathbf{B}\times\mathbf{C}$$

11. Evaluate $\dfrac{d}{dt}(\mathbf{V}\cdot\dfrac{d\mathbf{V}}{dt}\times\dfrac{d^2\mathbf{V}}{dt^2})$.

By Problem 10, $\dfrac{d}{dt}(\mathbf{V}\cdot\dfrac{d\mathbf{V}}{dt}\times\dfrac{d^2\mathbf{V}}{dt^2}) = \mathbf{V}\cdot\dfrac{d\mathbf{V}}{dt}\times\dfrac{d^3\mathbf{V}}{dt^3} + \mathbf{V}\cdot\dfrac{d^2\mathbf{V}}{dt^2}\times\dfrac{d^2\mathbf{V}}{dt^2} + \dfrac{d\mathbf{V}}{dt}\cdot\dfrac{d\mathbf{V}}{dt}\times\dfrac{d^2\mathbf{V}}{dt^2}$

$$= \mathbf{V}\cdot\frac{d\mathbf{V}}{dt}\times\frac{d^3\mathbf{V}}{dt^3} + 0 + 0 = \mathbf{V}\cdot\frac{d\mathbf{V}}{dt}\times\frac{d^3\mathbf{V}}{dt^3}$$

12. A particle moves so that its position vector is given by $\mathbf{r} = \cos\omega t\,\mathbf{i} + \sin\omega t\,\mathbf{j}$ where ω is a constant. Show that (*a*) the velocity $\mathbf{v}$ of the particle is perpendicular to $\mathbf{r}$, (*b*) the acceleration $\mathbf{a}$ is directed toward the origin and has magnitude proportional to the distance from the origin, (*c*) $\mathbf{r}\times\mathbf{v}$ = a constant vector.

(*a*) $\mathbf{v} = \frac{d\mathbf{r}}{dt} = -\omega \sin \omega t\, \mathbf{i} + \omega \cos \omega t\, \mathbf{j}$

Then $\mathbf{r}\cdot\mathbf{v} = [\cos \omega t\, \mathbf{i} + \sin \omega t\, \mathbf{j}] \cdot [-\omega \sin \omega t\, \mathbf{i} + \omega \cos \omega t\, \mathbf{j}]$

$= (\cos \omega t)(-\omega \sin \omega t) + (\sin \omega t)(\omega \cos \omega t) = 0$

and $\mathbf{r}$ and $\mathbf{v}$ are perpendicular.

(*b*) $\frac{d^2\mathbf{r}}{dt^2} = \frac{d\mathbf{v}}{dt} = -\omega^2 \cos \omega t\, \mathbf{i} - \omega^2 \sin \omega t\, \mathbf{j}$

$= -\omega^2 [\cos \omega t\, \mathbf{i} + \sin \omega t\, \mathbf{j}] = -\omega^2 \mathbf{r}$

Then the acceleration is opposite to the direction of $\mathbf{r}$, i.e. it is directed toward the origin. Its magnitude is proportional to $|\mathbf{r}|$ which is the distance from the origin.

(*c*) $\mathbf{r}\times\mathbf{v} = [\cos \omega t\, \mathbf{i} + \sin \omega t\, \mathbf{j}] \times [-\omega \sin \omega t\, \mathbf{i} + \omega \cos \omega t\, \mathbf{j}]$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos \omega t & \sin \omega t & 0 \\ -\omega \sin \omega t & \omega \cos \omega t & 0 \end{vmatrix} = \omega(\cos^2 \omega t + \sin^2 \omega t)\mathbf{k} = \omega \mathbf{k}, \text{ a constant vector.}$$

Physically, the motion is that of a particle moving on the circumference of a circle with constant angular speed ω. The acceleration, directed toward the center of the circle, is the *centripetal acceleration.*

13. Prove: $\mathbf{A}\times\frac{d^2\mathbf{B}}{dt^2} - \frac{d^2\mathbf{A}}{dt^2}\times\mathbf{B} = \frac{d}{dt}(\mathbf{A}\times\frac{d\mathbf{B}}{dt} - \frac{d\mathbf{A}}{dt}\times\mathbf{B})$.

$$\frac{d}{dt}(\mathbf{A}\times\frac{d\mathbf{B}}{dt} - \frac{d\mathbf{A}}{dt}\times\mathbf{B}) = \frac{d}{dt}(\mathbf{A}\times\frac{d\mathbf{B}}{dt}) - \frac{d}{dt}(\frac{d\mathbf{A}}{dt}\times\mathbf{B})$$

$$= \mathbf{A}\times\frac{d^2\mathbf{B}}{dt^2} + \frac{d\mathbf{A}}{dt}\times\frac{d\mathbf{B}}{dt} - [\frac{d\mathbf{A}}{dt}\times\frac{d\mathbf{B}}{dt} + \frac{d^2\mathbf{A}}{dt^2}\times\mathbf{B}] = \mathbf{A}\times\frac{d^2\mathbf{B}}{dt^2} - \frac{d^2\mathbf{A}}{dt^2}\times\mathbf{B}$$

14. Show that $\mathbf{A}\cdot\frac{d\mathbf{A}}{dt} = A\frac{dA}{dt}$.

Let $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$. Then $A = \sqrt{A_1^2 + A_2^2 + A_3^2}$.

$$\frac{dA}{dt} = \frac{1}{2}(A_1^2 + A_2^2 + A_3^2)^{-1/2}(2A_1\frac{dA_1}{dt} + 2A_2\frac{dA_2}{dt} + 2A_3\frac{dA_3}{dt})$$

$$= \frac{A_1\frac{dA_1}{dt} + A_2\frac{dA_2}{dt} + A_3\frac{dA_3}{dt}}{(A_1^2 + A_2^2 + A_3^2)^{1/2}} = \frac{\mathbf{A}\cdot\frac{d\mathbf{A}}{dt}}{A}, \quad \text{i.e. } A\frac{dA}{dt} = \mathbf{A}\cdot\frac{d\mathbf{A}}{dt}.$$

Another Method.

Since $\mathbf{A}\cdot\mathbf{A} = A^2$, $\frac{d}{dt}(\mathbf{A}\cdot\mathbf{A}) = \frac{d}{dt}(A^2)$.

$$\frac{d}{dt}(\mathbf{A}\cdot\mathbf{A}) = \mathbf{A}\cdot\frac{d\mathbf{A}}{dt} + \frac{d\mathbf{A}}{dt}\cdot\mathbf{A} = 2\mathbf{A}\cdot\frac{d\mathbf{A}}{dt} \quad \text{and} \quad \frac{d}{dt}(A^2) = 2A\frac{dA}{dt}$$

Then $2\mathbf{A}\cdot\frac{d\mathbf{A}}{dt} = 2A\frac{dA}{dt}$ or $\mathbf{A}\cdot\frac{d\mathbf{A}}{dt} = A\frac{dA}{dt}$.

Note that if $\mathbf{A}$ is a constant vector $\mathbf{A}\cdot\frac{d\mathbf{A}}{dt} = 0$ as in Problem 9.

15. If $\mathbf{A} = (2x^2y - x^4)\mathbf{i} + (e^{xy} - y\sin x)\mathbf{j} + (x^2\cos y)\mathbf{k}$, find: $\dfrac{\partial \mathbf{A}}{\partial x}, \dfrac{\partial \mathbf{A}}{\partial y}, \dfrac{\partial^2 \mathbf{A}}{\partial x^2}, \dfrac{\partial^2 \mathbf{A}}{\partial y^2}, \dfrac{\partial^2 \mathbf{A}}{\partial x\,\partial y}, \dfrac{\partial^2 \mathbf{A}}{\partial y\,\partial x}$.

$$\frac{\partial \mathbf{A}}{\partial x} = \frac{\partial}{\partial x}(2x^2y - x^4)\mathbf{i} + \frac{\partial}{\partial x}(e^{xy} - y\sin x)\mathbf{j} + \frac{\partial}{\partial x}(x^2\cos y)\mathbf{k}$$

$$= (4xy - 4x^3)\mathbf{i} + (ye^{xy} - y\cos x)\mathbf{j} + 2x\cos y\,\mathbf{k}$$

$$\frac{\partial \mathbf{A}}{\partial y} = \frac{\partial}{\partial y}(2x^2y - x^4)\mathbf{i} + \frac{\partial}{\partial y}(e^{xy} - y\sin x)\mathbf{j} + \frac{\partial}{\partial y}(x^2\cos y)\mathbf{k}$$

$$= 2x^2\,\mathbf{i} + (xe^{xy} - \sin x)\mathbf{j} - x^2\sin y\,\mathbf{k}$$

$$\frac{\partial^2 \mathbf{A}}{\partial x^2} = \frac{\partial}{\partial x}(4xy - 4x^3)\mathbf{i} + \frac{\partial}{\partial x}(ye^{xy} - y\cos x)\mathbf{j} + \frac{\partial}{\partial x}(2x\cos y)\mathbf{k}$$

$$= (4y - 12x^2)\mathbf{i} + (y^2e^{xy} + y\sin x)\mathbf{j} + 2\cos y\,\mathbf{k}$$

$$\frac{\partial^2 \mathbf{A}}{\partial y^2} = \frac{\partial}{\partial y}(2x^2)\mathbf{i} + \frac{\partial}{\partial y}(xe^{xy} - \sin x)\mathbf{j} - \frac{\partial}{\partial y}(x^2\sin y)\mathbf{k}$$

$$= \mathbf{0} + x^2e^{xy}\,\mathbf{j} - x^2\cos y\,\mathbf{k} = x^2e^{xy}\,\mathbf{j} - x^2\cos y\,\mathbf{k}$$

$$\frac{\partial^2 \mathbf{A}}{\partial x\,\partial y} = \frac{\partial}{\partial x}\left(\frac{\partial \mathbf{A}}{\partial y}\right) = \frac{\partial}{\partial x}(2x^2)\mathbf{i} + \frac{\partial}{\partial x}(xe^{xy} - \sin x)\mathbf{j} - \frac{\partial}{\partial x}(x^2\sin y)\mathbf{k}$$

$$= 4x\,\mathbf{i} + (xye^{xy} + e^{xy} - \cos x)\mathbf{j} - 2x\sin y\,\mathbf{k}$$

$$\frac{\partial^2 \mathbf{A}}{\partial y\,\partial x} = \frac{\partial}{\partial y}\left(\frac{\partial \mathbf{A}}{\partial x}\right) = \frac{\partial}{\partial y}(4xy - 4x^3)\mathbf{i} + \frac{\partial}{\partial y}(ye^{xy} - y\cos x)\mathbf{j} + \frac{\partial}{\partial y}(2x\cos y)\mathbf{k}$$

$$= 4x\,\mathbf{i} + (xye^{xy} + e^{xy} - \cos x)\mathbf{j} - 2x\sin y\,\mathbf{k}$$

Note that $\dfrac{\partial^2 \mathbf{A}}{\partial y\,\partial x} = \dfrac{\partial^2 \mathbf{A}}{\partial x\,\partial y}$, i.e. the order of differentiation is immaterial. This is true in general if $\mathbf{A}$ has continuous partial derivatives of the second order at least.

16. If $\phi(x,y,z) = xy^2z$ and $\mathbf{A} = xz\,\mathbf{i} - xy^2\,\mathbf{j} + yz^2\mathbf{k}$, find $\dfrac{\partial^3}{\partial x^2\,\partial z}(\phi\mathbf{A})$ at the point $(2,-1,1)$.

$$\phi\mathbf{A} = (xy^2z)(xz\,\mathbf{i} - xy^2\,\mathbf{j} + yz^2\,\mathbf{k}) = x^2y^2z^2\,\mathbf{i} - x^2y^4z\,\mathbf{j} + xy^3z^3\,\mathbf{k}$$

$$\frac{\partial}{\partial z}(\phi\mathbf{A}) = \frac{\partial}{\partial z}(x^2y^2z^2\,\mathbf{i} - x^2y^4z\,\mathbf{j} + xy^3z^3\,\mathbf{k}) = 2x^2y^2z\,\mathbf{i} - x^2y^4\,\mathbf{j} + 3xy^3z^2\,\mathbf{k}$$

$$\frac{\partial^2}{\partial x\,\partial z}(\phi\mathbf{A}) = \frac{\partial}{\partial x}(2x^2y^2z\,\mathbf{i} - x^2y^4\,\mathbf{j} + 3xy^3z^2\,\mathbf{k}) = 4xy^2z\,\mathbf{i} - 2xy^4\,\mathbf{j} + 3y^3z^2\,\mathbf{k}$$

$$\frac{\partial^3}{\partial x^2\,\partial z}(\phi\mathbf{A}) = \frac{\partial}{\partial x}(4xy^2z\,\mathbf{i} - 2xy^4\,\mathbf{j} + 3y^3z^2\,\mathbf{k}) = 4y^2z\,\mathbf{i} - 2y^4\,\mathbf{j}$$

If $x=2$, $y=-1$, $z=1$ this becomes $4(-1)^2(1)\mathbf{i} - 2(-1)^4\,\mathbf{j} = 4\mathbf{i} - 2\mathbf{j}$.

17. Let $\mathbf{F}$ depend on x, y, z, t where x, y and z depend on t. Prove that

$$\frac{d\mathbf{F}}{dt} = \frac{\partial \mathbf{F}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x}\frac{dx}{dt} + \frac{\partial \mathbf{F}}{\partial y}\frac{dy}{dt} + \frac{\partial \mathbf{F}}{\partial z}\frac{dz}{dt}$$

under suitable assumptions of differentiability.

Suppose that $\mathbf{F} = F_1(x,y,z,t)\mathbf{i} + F_2(x,y,z,t)\mathbf{j} + F_3(x,y,z,t)\mathbf{k}$. Then

$$\begin{aligned} d\mathbf{F} &= dF_1\,\mathbf{i} + dF_2\,\mathbf{j} + dF_3\,\mathbf{k} \\ &= [\frac{\partial F_1}{\partial t}dt + \frac{\partial F_1}{\partial x}dx + \frac{\partial F_1}{\partial y}dy + \frac{\partial F_1}{\partial z}dz]\mathbf{i} + [\frac{\partial F_2}{\partial t}dt + \frac{\partial F_2}{\partial x}dx + \frac{\partial F_2}{\partial y}dy + \frac{\partial F_2}{\partial z}dz]\mathbf{j} \\ &\quad + [\frac{\partial F_3}{\partial t}dt + \frac{\partial F_3}{\partial x}dx + \frac{\partial F_3}{\partial y}dy + \frac{\partial F_3}{\partial z}dz]\mathbf{k} \\ &= (\frac{\partial F_1}{\partial t}\mathbf{i} + \frac{\partial F_2}{\partial t}\mathbf{j} + \frac{\partial F_3}{\partial t}\mathbf{k})dt + (\frac{\partial F_1}{\partial x}\mathbf{i} + \frac{\partial F_2}{\partial x}\mathbf{j} + \frac{\partial F_3}{\partial x}\mathbf{k})dx \\ &\quad + (\frac{\partial F_1}{\partial y}\mathbf{i} + \frac{\partial F_2}{\partial y}\mathbf{j} + \frac{\partial F_3}{\partial y}\mathbf{k})dy + (\frac{\partial F_1}{\partial z}\mathbf{i} + \frac{\partial F_2}{\partial z}\mathbf{j} + \frac{\partial F_3}{\partial z}\mathbf{k})dz \\ &= \frac{\partial \mathbf{F}}{\partial t}dt + \frac{\partial \mathbf{F}}{\partial x}dx + \frac{\partial \mathbf{F}}{\partial y}dy + \frac{\partial \mathbf{F}}{\partial z}dz \end{aligned}$$

and so $$\frac{d\mathbf{F}}{dt} = \frac{\partial \mathbf{F}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x}\frac{dx}{dt} + \frac{\partial \mathbf{F}}{\partial y}\frac{dy}{dt} + \frac{\partial \mathbf{F}}{\partial z}\frac{dz}{dt}.$$

DIFFERENTIAL GEOMETRY.

18. Prove the Frenet-Serret formulas (*a*) $\frac{d\mathbf{T}}{ds} = \kappa\mathbf{N}$, (*b*) $\frac{d\mathbf{B}}{ds} = -\tau\mathbf{N}$, (*c*) $\frac{d\mathbf{N}}{ds} = \tau\mathbf{B} - \kappa\mathbf{T}$.

(*a*) Since $\mathbf{T}\cdot\mathbf{T} = 1$, it follows from Problem 9 that $\mathbf{T}\cdot\frac{d\mathbf{T}}{ds} = 0$, i.e. $\frac{d\mathbf{T}}{ds}$ is perpendicular to $\mathbf{T}$.

If $\mathbf{N}$ is a unit vector in the direction $\frac{d\mathbf{T}}{ds}$, then $\frac{d\mathbf{T}}{ds} = \kappa\mathbf{N}$. We call $\mathbf{N}$ the *principal normal*, κ the *curvature* and $\rho = 1/\kappa$ the *radius of curvature*.

(*b*) Let $\mathbf{B} = \mathbf{T}\times\mathbf{N}$, so that $\frac{d\mathbf{B}}{ds} = \mathbf{T}\times\frac{d\mathbf{N}}{ds} + \frac{d\mathbf{T}}{ds}\times\mathbf{N} = \mathbf{T}\times\frac{d\mathbf{N}}{ds} + \kappa\mathbf{N}\times\mathbf{N} = \mathbf{T}\times\frac{d\mathbf{N}}{ds}$.

Then $\mathbf{T}\cdot\frac{d\mathbf{B}}{ds} = \mathbf{T}\cdot\mathbf{T}\times\frac{d\mathbf{N}}{ds} = 0$, so that $\mathbf{T}$ is perpendicular to $\frac{d\mathbf{B}}{ds}$.

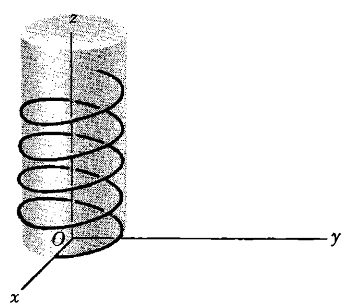
But from $\mathbf{B}\cdot\mathbf{B} = 1$ it follows that $\mathbf{B}\cdot\frac{d\mathbf{B}}{ds} = 0$ (Problem 9), so that $\frac{d\mathbf{B}}{ds}$ is perpendicular to $\mathbf{B}$ and is thus in the plane of $\mathbf{T}$ and $\mathbf{N}$.

Since $\frac{d\mathbf{B}}{ds}$ is in the plane of $\mathbf{T}$ and $\mathbf{N}$ and is perpendicular to $\mathbf{T}$, it must be parallel to $\mathbf{N}$; then $\frac{d\mathbf{B}}{ds} = -\tau\mathbf{N}$. We call $\mathbf{B}$ the *binormal*, τ the *torsion*, and $\sigma = 1/\tau$ the *radius of torsion*.

(*c*) Since $\mathbf{T}, \mathbf{N}, \mathbf{B}$ form a right-handed system, so do $\mathbf{N}, \mathbf{B}$ and $\mathbf{T}$, i.e. $\mathbf{N} = \mathbf{B}\times\mathbf{T}$.

Then $\frac{d\mathbf{N}}{ds} = \mathbf{B}\times\frac{d\mathbf{T}}{ds} + \frac{d\mathbf{B}}{ds}\times\mathbf{T} = \mathbf{B}\times\kappa\mathbf{N} - \tau\mathbf{N}\times\mathbf{T} = -\kappa\mathbf{T} + \tau\mathbf{B} = \tau\mathbf{B} - \kappa\mathbf{T}$.

19. Sketch the space curve $x = 3\cos t$, $y = 3\sin t$, $z = 4t$ and find (*a*) the unit tangent $\mathbf{T}$, (*b*) the principal normal $\mathbf{N}$, curvature κ and radius of curvature ρ, (*c*) the binormal $\mathbf{B}$, torsion τ and radius of torsion σ.



The space curve is a *circular helix* (see adjacent figure). Since $t = z/4$, the curve has equations $x = 3\cos(z/4)$, $y = 3\sin(z/4)$ and therefore lies on the cylinder $x^2 + y^2 = 9$.

(*a*) The position vector for any point on the curve is

$$\mathbf{r} = 3\cos t\,\mathbf{i} + 3\sin t\,\mathbf{j} + 4t\,\mathbf{k}$$

Then $\dfrac{d\mathbf{r}}{dt} = -3\sin t\,\mathbf{i} + 3\cos t\,\mathbf{j} + 4\mathbf{k}$

$$\frac{ds}{dt} = \left|\frac{d\mathbf{r}}{dt}\right| = \sqrt{\frac{d\mathbf{r}}{dt}\cdot\frac{d\mathbf{r}}{dt}} = \sqrt{(-3\sin t)^2 + (3\cos t)^2 + 4^2} = 5$$

Thus $\mathbf{T} = \dfrac{d\mathbf{r}}{ds} = \dfrac{d\mathbf{r}/dt}{ds/dt} = -\dfrac{3}{5}\sin t\,\mathbf{i} + \dfrac{3}{5}\cos t\,\mathbf{j} + \dfrac{4}{5}\mathbf{k}$.

(b) $\dfrac{d\mathbf{T}}{dt} = \dfrac{d}{dt}(-\dfrac{3}{5}\sin t\,\mathbf{i} + \dfrac{3}{5}\cos t\,\mathbf{j} + \dfrac{4}{5}\mathbf{k}) = -\dfrac{3}{5}\cos t\,\mathbf{i} - \dfrac{3}{5}\sin t\,\mathbf{j}$

$$\frac{d\mathbf{T}}{ds} = \frac{d\mathbf{T}/dt}{ds/dt} = -\frac{3}{25}\cos t\,\mathbf{i} - \frac{3}{25}\sin t\,\mathbf{j}$$

Since $\dfrac{d\mathbf{T}}{ds} = \kappa\mathbf{N}$, $\left|\dfrac{d\mathbf{T}}{ds}\right| = |\kappa|\,|\mathbf{N}| = \kappa$ as $\kappa \geqq 0$.

Then $\kappa = \left|\dfrac{d\mathbf{T}}{ds}\right| = \sqrt{(-\dfrac{3}{25}\cos t)^2 + (-\dfrac{3}{25}\sin t)^2} = \dfrac{3}{25}$ and $\rho = \dfrac{1}{\kappa} = \dfrac{25}{3}$.

From $\dfrac{d\mathbf{T}}{ds} = \kappa\mathbf{N}$, we obtain $\mathbf{N} = \dfrac{1}{\kappa}\dfrac{d\mathbf{T}}{ds} = -\cos t\,\mathbf{i} - \sin t\,\mathbf{j}$.

(c) $\mathbf{B} = \mathbf{T}\times\mathbf{N} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\frac{3}{5}\sin t & \frac{3}{5}\cos t & \frac{4}{5} \\ -\cos t & -\sin t & 0 \end{vmatrix} = \dfrac{4}{5}\sin t\,\mathbf{i} - \dfrac{4}{5}\cos t\,\mathbf{j} + \dfrac{3}{5}\mathbf{k}$

$\dfrac{d\mathbf{B}}{dt} = \dfrac{4}{5}\cos t\,\mathbf{i} + \dfrac{4}{5}\sin t\,\mathbf{j}$, $\dfrac{d\mathbf{B}}{ds} = \dfrac{d\mathbf{B}/dt}{ds/dt} = \dfrac{4}{25}\cos t\,\mathbf{i} + \dfrac{4}{25}\sin t\,\mathbf{j}$

$-\tau\mathbf{N} = -\tau(-\cos t\,\mathbf{i} - \sin t\,\mathbf{j}) = \dfrac{4}{25}\cos t\,\mathbf{i} + \dfrac{4}{25}\sin t\,\mathbf{j}$ or $\tau = \dfrac{4}{25}$ and $\sigma = \dfrac{1}{\tau} = \dfrac{25}{4}$

20. Prove that the radius of curvature of the curve with parametric equations $x = x(s)$, $y = y(s)$, $z = z(s)$ is given by $\rho = \left[(\dfrac{d^2x}{ds^2})^2 + (\dfrac{d^2y}{ds^2})^2 + (\dfrac{d^2z}{ds^2})^2\right]^{-1/2}$.

The position vector of any point on the curve is $\mathbf{r} = x(s)\mathbf{i} + y(s)\mathbf{j} + z(s)\mathbf{k}$.

Then $\mathbf{T} = \dfrac{d\mathbf{r}}{ds} = \dfrac{dx}{ds}\mathbf{i} + \dfrac{dy}{ds}\mathbf{j} + \dfrac{dz}{ds}\mathbf{k}$ and $\dfrac{d\mathbf{T}}{ds} = \dfrac{d^2x}{ds^2}\mathbf{i} + \dfrac{d^2y}{ds^2}\mathbf{j} + \dfrac{d^2z}{ds^2}\mathbf{k}$.

But $\dfrac{d\mathbf{T}}{ds} = \kappa\mathbf{N}$ so that $\kappa = \left|\dfrac{d\mathbf{T}}{ds}\right| = \sqrt{(\dfrac{d^2x}{ds^2})^2 + (\dfrac{d^2y}{ds^2})^2 + (\dfrac{d^2z}{ds^2})^2}$ and the result follows since $\rho = \dfrac{1}{\kappa}$.

21. Show that $\dfrac{d\mathbf{r}}{ds}\cdot\dfrac{d^2\mathbf{r}}{ds^2}\times\dfrac{d^3\mathbf{r}}{ds^3} = \dfrac{\tau}{\rho^2}$.

$\dfrac{d\mathbf{r}}{ds} = \mathbf{T}$, $\dfrac{d^2\mathbf{r}}{ds^2} = \dfrac{d\mathbf{T}}{ds} = \kappa\mathbf{N}$, $\dfrac{d^3\mathbf{r}}{ds^3} = \kappa\dfrac{d\mathbf{N}}{ds} + \dfrac{d\kappa}{ds}\mathbf{N} = \kappa(\tau\mathbf{B} - \kappa\mathbf{T}) + \dfrac{d\kappa}{ds}\mathbf{N} = \kappa\tau\mathbf{B} - \kappa^2\mathbf{T} + \dfrac{d\kappa}{ds}\mathbf{N}$

$$\frac{d\mathbf{r}}{ds}\cdot\frac{d^2\mathbf{r}}{ds^2}\times\frac{d^3\mathbf{r}}{ds^3} = \mathbf{T}\cdot\kappa\mathbf{N}\times(\kappa\tau\mathbf{B} - \kappa^2\mathbf{T} + \frac{d\kappa}{ds}\mathbf{N})$$

$$= \mathbf{T}\cdot(\kappa^2\tau\,\mathbf{N}\times\mathbf{B} - \kappa^3\mathbf{N}\times\mathbf{T} + \kappa\frac{d\kappa}{ds}\mathbf{N}\times\mathbf{N}) = \mathbf{T}\cdot(\kappa^2\tau\,\mathbf{T} + \kappa^3\mathbf{B}) = \kappa^2\tau = \frac{\tau}{\rho^2}$$

The result can be written

$$\tau = [(x'')^2 + (y'')^2 + (z'')^2]^{-1} \begin{vmatrix} x' & y' & z' \\ x'' & y'' & z'' \\ x''' & y''' & z''' \end{vmatrix}$$

where primes denote derivatives with respect to s, by using the result of Problem 20.

22. Given the space curve $x = t,\ y = t^2,\ z = \frac{2}{3}t^3$, find (*a*) the curvature κ, (*b*) the torsion τ.

(*a*) The position vector is $\mathbf{r} = t\,\mathbf{i} + t^2\,\mathbf{j} + \frac{2}{3}t^3\,\mathbf{k}$.

Then $\dfrac{d\mathbf{r}}{dt} = \mathbf{i} + 2t\,\mathbf{j} + 2t^2\,\mathbf{k}$

$$\frac{ds}{dt} = \left|\frac{d\mathbf{r}}{dt}\right| = \sqrt{\frac{d\mathbf{r}}{dt}\cdot\frac{d\mathbf{r}}{dt}} = \sqrt{(1)^2 + (2t)^2 + (2t^2)^2} = 1 + 2t^2$$

and $\mathbf{T} = \dfrac{d\mathbf{r}}{ds} = \dfrac{d\mathbf{r}/dt}{ds/dt} = \dfrac{\mathbf{i} + 2t\,\mathbf{j} + 2t^2\,\mathbf{k}}{1 + 2t^2}$.

$$\frac{d\mathbf{T}}{dt} = \frac{(1+2t^2)(2\mathbf{j} + 4t\,\mathbf{k}) - (\mathbf{i} + 2t\,\mathbf{j} + 2t^2\,\mathbf{k})(4t)}{(1+2t^2)^2} = \frac{-4t\,\mathbf{i} + (2 - 4t^2)\mathbf{j} + 4t\,\mathbf{k}}{(1+2t^2)^2}$$

Then $\dfrac{d\mathbf{T}}{ds} = \dfrac{d\mathbf{T}/dt}{ds/dt} = \dfrac{-4t\,\mathbf{i} + (2-4t^2)\mathbf{j} + 4t\,\mathbf{k}}{(1+2t^2)^3}$.

Since $\dfrac{d\mathbf{T}}{ds} = \kappa\mathbf{N}$, $\kappa = \left|\dfrac{d\mathbf{T}}{ds}\right| = \dfrac{\sqrt{(-4t)^2 + (2-4t^2)^2 + (4t)^2}}{(1+2t^2)^3} = \dfrac{2}{(1+2t^2)^2}$

(*b*) From (*a*), $\mathbf{N} = \dfrac{1}{\kappa}\dfrac{d\mathbf{T}}{ds} = \dfrac{-2t\,\mathbf{i} + (1-2t^2)\mathbf{j} + 2t\,\mathbf{k}}{1+2t^2}$

Then $\mathbf{B} = \mathbf{T}\times\mathbf{N} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{1}{1+2t^2} & \dfrac{2t}{1+2t^2} & \dfrac{2t^2}{1+2t^2} \\ \dfrac{-2t}{1+2t^2} & \dfrac{1-2t^2}{1+2t^2} & \dfrac{2t}{1+2t^2} \end{vmatrix} = \dfrac{2t^2\,\mathbf{i} - 2t\,\mathbf{j} + \mathbf{k}}{1+2t^2}$

Now $\dfrac{d\mathbf{B}}{dt} = \dfrac{4t\,\mathbf{i} + (4t^2-2)\mathbf{j} - 4t\,\mathbf{k}}{(1+2t^2)^2}$ and $\dfrac{d\mathbf{B}}{ds} = \dfrac{d\mathbf{B}/dt}{ds/dt} = \dfrac{4t\,\mathbf{i} + (4t^2-2)\mathbf{j} - 4t\,\mathbf{k}}{(1+2t^2)^3}$

Also, $-\tau\mathbf{N} = -\tau\left[\dfrac{-2t\,\mathbf{i} + (1-2t^2)\mathbf{j} + 2t\,\mathbf{k}}{1+2t^2}\right]$. Since $\dfrac{d\mathbf{B}}{ds} = -\tau\mathbf{N}$, we find $\tau = \dfrac{2}{(1+2t^2)^2}$.

Note that $\kappa = \tau$ for this curve.

23. Find equations in vector and rectangular form for the (*a*) tangent, (*b*) principal normal, and (*c*) binormal to the curve of Problem 22 at the point where $t = 1$.

Let $\mathbf{T}_0, \mathbf{N}_0$ and $\mathbf{B}_0$ denote the tangent, principal normal and binormal vectors at the required point. Then from Problem 22,

$$\mathbf{T}_0 = \frac{\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}}{3}, \quad \mathbf{N}_0 = \frac{-2\mathbf{i} - \mathbf{j} + 2\mathbf{k}}{3}, \quad \mathbf{B}_0 = \frac{2\mathbf{i} - 2\mathbf{j} + \mathbf{k}}{3}$$

If $\mathbf{A}$ denotes a given vector while $\mathbf{r}_0$ and $\mathbf{r}$ denote respectively the position vectors of the initial point and an arbitrary point of $\mathbf{A}$, then $\mathbf{r}-\mathbf{r}_0$ is parallel to $\mathbf{A}$ and so the equation of $\mathbf{A}$ is $(\mathbf{r}-\mathbf{r}_0)\times\mathbf{A} = \mathbf{0}$.

Then:

	Equation of tangent is	$(\mathbf{r}-\mathbf{r}_0)\times\mathbf{T}_0 = \mathbf{0}$
Then:	Equation of principal normal is	$(\mathbf{r}-\mathbf{r}_0)\times\mathbf{N}_0 = \mathbf{0}$
	Equation of binormal is	$(\mathbf{r}-\mathbf{r}_0)\times\mathbf{B}_0 = \mathbf{0}$

In rectangular form, with $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, $\mathbf{r}_0 = \mathbf{i} + \mathbf{j} + \frac{2}{3}\mathbf{k}$ these become respectively

$$\frac{x-1}{1} = \frac{y-1}{2} = \frac{z-2/3}{2}, \quad \frac{x-1}{-2} = \frac{y-1}{-1} = \frac{z-2/3}{2}, \quad \frac{x-1}{2} = \frac{y-1}{-2} = \frac{z-2/3}{1}.$$

These equations can also be written in parametric form (see Problem 28, Chapter 1).

24. Find equations in vector and rectangular form for the (*a*) osculating plane, (*b*) normal plane, and (*c*) rectifying plane to the curve of Problems 22 and 23 at the point where $t = 1$.

(*a*) The osculating plane is the plane which contains the tangent and principal normal. If $\mathbf{r}$ is the position vector of any point in this plane and $\mathbf{r}_0$ is the position vector of the point $t=1$, then $\mathbf{r}-\mathbf{r}_0$ is perpendicular to $\mathbf{B}_0$, the binormal at the point $t=1$, i.e. $(\mathbf{r}-\mathbf{r}_0)\cdot\mathbf{B}_0 = 0$.

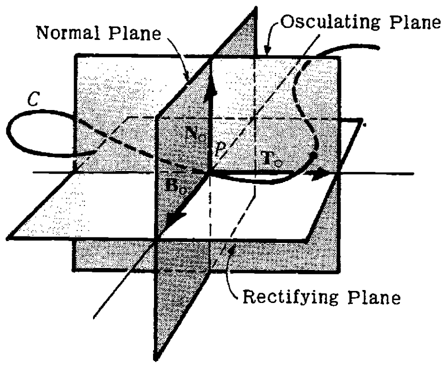
(*b*) The normal plane is the plane which is perpendicular to the tangent vector at the given point. Then the required equation is $(\mathbf{r}-\mathbf{r}_0)\cdot\mathbf{T}_0 = 0$.

(*c*) The rectifying plane is the plane which is perpendicular to the principal normal at the given point. The required equation is $(\mathbf{r}-\mathbf{r}_0)\cdot\mathbf{N}_0 = 0$.

In rectangular form the equations of (*a*), (*b*) and (*c*) become respectively,

$$2(x-1) - 2(y-1) + 1(z-2/3) = 0,$$
$$1(x-1) + 2(y-1) + 2(z-2/3) = 0,$$
$$-2(x-1) - 1(y-1) + 2(z-2/3) = 0.$$

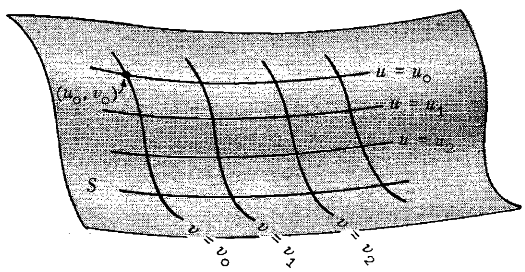
The adjoining figure shows the osculating, normal and rectifying planes to a curve C at the point P.



25. (*a*) Show that the equation $\mathbf{r} = \mathbf{r}(u, v)$ represents a surface.
(*b*) Show that $\dfrac{\partial\mathbf{r}}{\partial u}\times\dfrac{\partial\mathbf{r}}{\partial v}$ represents a vector normal to the surface.
(*c*) Determine a unit normal to the following surface, where $a > 0$:

$$\mathbf{r} = a\cos u\,\sin v\,\mathbf{i} + a\sin u\,\sin v\,\mathbf{j} + a\cos v\,\mathbf{k}$$

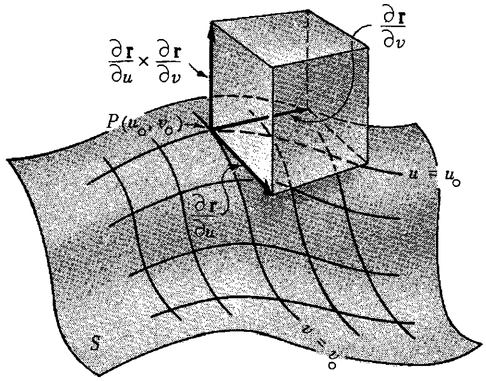
(*a*) If we consider u to have a fixed value, say u_0, then $\mathbf{r} = \mathbf{r}(u_0, v)$ represents a curve which can be denoted by $u = u_0$. Similarly $u = u_1$ defines another curve $\mathbf{r} = \mathbf{r}(u_1, v)$. As u varies, therefore, $\mathbf{r} = \mathbf{r}(u,v)$ represents a curve which moves in space and generates a surface S. Then $\mathbf{r} = \mathbf{r}(u,v)$ represents the surface S thus generated, as shown in the adjoining figure.



The curves $u = u_0$, $u = u_1$, ... represent definite curves on the surface. Similarly $v = v_0$, $v = v_1$, ... represent curves on the surface.

By assigning definite values to u and v, we obtain a point on the surface. Thus curves $u = u_0$ and $v = v_0$, for example, intersect and define the point (u_0, v_0) on the surface. We speak of the pair of numbers (u,v) as defining the *curvilinear coordinates* on the surface. If all the curves $u =$ constant and $v =$ constant are perpendicular at each point of intersection, we call the curvilinear coordinate system *orthogonal*. For further discussion of curvilinear coordinates see Chapter 7.

(b) Consider point P having coordinates (u_0, v_0) on a surface S, as shown in the adjacent diagram. The vector $\partial\mathbf{r}/\partial u$ at P is obtained by differentiating $\mathbf{r}$ with respect to u, keeping $v = \text{constant} = v_0$. From the theory of space curves, it follows that $\partial\mathbf{r}/\partial u$ at P represents a vector tangent to the curve $v = v_0$ at P, as shown in the adjoining figure. Similarly, $\partial\mathbf{r}/\partial v$ at P represents a vector tangent to the curve $u = \text{constant} = u_0$. Since $\partial\mathbf{r}/\partial u$ and $\partial\mathbf{r}/\partial v$ represent vectors at P tangent to curves which lie on the surface S at P, it follows that these vectors are tangent to the surface at P. Hence it follows that $\dfrac{\partial\mathbf{r}}{\partial u}\times\dfrac{\partial\mathbf{r}}{\partial v}$ is a vector normal to S at P.



(c) $$\frac{\partial\mathbf{r}}{\partial u} = -a\sin u\,\sin v\,\mathbf{i} + a\cos u\,\sin v\,\mathbf{j}$$

$$\frac{\partial\mathbf{r}}{\partial v} = a\cos u\,\cos v\,\mathbf{i} + a\sin u\,\cos v\,\mathbf{j} - a\sin v\,\mathbf{k}$$

Then $$\frac{\partial\mathbf{r}}{\partial u}\times\frac{\partial\mathbf{r}}{\partial v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -a\sin u\,\sin v & a\cos u\,\sin v & 0 \\ a\cos u\,\cos v & a\sin u\,\cos v & -a\sin v \end{vmatrix}$$

$$= -a^2\cos u\,\sin^2 v\,\mathbf{i} - a^2\sin u\,\sin^2 v\,\mathbf{j} - a^2\sin v\,\cos v\,\mathbf{k}$$

represents a vector normal to the surface at any point (u,v).

A unit normal is obtained by dividing $\dfrac{\partial\mathbf{r}}{\partial u}\times\dfrac{\partial\mathbf{r}}{\partial v}$ by its magnitude, $\left|\dfrac{\partial\mathbf{r}}{\partial u}\times\dfrac{\partial\mathbf{r}}{\partial v}\right|$, given by

$$\sqrt{a^4\cos^2 u\,\sin^4 v + a^4\sin^2 u\,\sin^4 v + a^4\sin^2 v\,\cos^2 v}$$
$$= \sqrt{a^4(\cos^2 u + \sin^2 u)\sin^4 v + a^4\sin^2 v\,\cos^2 v}$$
$$= \sqrt{a^4\sin^2 v(\sin^2 v + \cos^2 v)} = \begin{cases} a^2\sin v & \text{if } \sin v > 0 \\ -a^2\sin v & \text{if } \sin v < 0 \end{cases}$$

Then there are two unit normals given by

$$\pm(\cos u\,\sin v\,\mathbf{i} + \sin u\,\sin v\,\mathbf{j} + \cos v\,\mathbf{k}) = \pm\mathbf{n}$$

It should be noted that the given surface is defined by $x = a\cos u\,\sin v$, $y = a\sin u\,\sin v$, $z = a\cos v$ from which it is seen that $x^2 + y^2 + z^2 = a^2$, which is a sphere of radius a. Since $\mathbf{r} = a\,\mathbf{n}$, it follows that

$$\mathbf{n} = \cos u\,\sin v\,\mathbf{i} + \sin u\,\sin v\,\mathbf{j} + \cos v\,\mathbf{k}$$

is the *outward drawn unit normal* to the sphere at the point (u,v).

26. Find an equation for the tangent plane to the surface $z = x^2 + y^2$ at the point $(1, -1, 2)$.

Let $x = u$, $y = v$, $z = u^2 + v^2$ be parametric equations of the surface. The position vector to any point on the surface is

$$\mathbf{r} = u\,\mathbf{i} + v\,\mathbf{j} + (u^2+v^2)\mathbf{k}$$

Then $\frac{\partial \mathbf{r}}{\partial u} = \mathbf{i} + 2u\,\mathbf{k} = \mathbf{i} + 2\mathbf{k}$, $\frac{\partial \mathbf{r}}{\partial v} = \mathbf{j} + 2v\,\mathbf{k} = \mathbf{j} - 2\mathbf{k}$ at the point $(1,-1,2)$, where $u=1$ and $v=-1$.

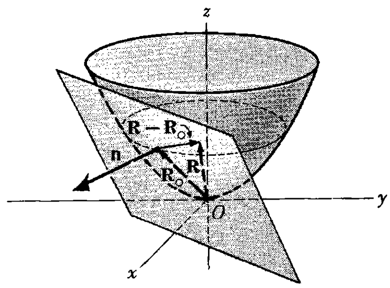
By Problem 25, a normal $\mathbf{n}$ to the surface at this point is

$$\mathbf{n} = \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} = (\mathbf{i} + 2\mathbf{k}) \times (\mathbf{j} - 2\mathbf{k}) = -2\mathbf{i} + 2\mathbf{j} + \mathbf{k}$$

The position vector to point $(1,-1,2)$ is $\mathbf{R}_0 = \mathbf{i} - \mathbf{j} + 2\mathbf{k}$. The position vector to any point on the plane is

$$\mathbf{R} = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}$$

Then from the adjoining figure, $\mathbf{R} - \mathbf{R}_0$ is perpendicular to $\mathbf{n}$ and the required equation of the plane is $(\mathbf{R} - \mathbf{R}_0)\cdot\mathbf{n} = 0$ or $[(x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}) - (\mathbf{i} - \mathbf{j} + 2\mathbf{k})]\cdot[-2\mathbf{i} + 2\mathbf{j} + \mathbf{k}] = 0$
i.e. $-2(x-1) + 2(y+1) + (z-2) = 0$ or $2x - 2y - z = 2$.



MECHANICS

27. Show that the acceleration $\mathbf{a}$ of a particle which travels along a space curve with velocity $\mathbf{v}$ is given by

$$\mathbf{a} = \frac{dv}{dt}\mathbf{T} + \frac{v^2}{\rho}\mathbf{N}$$

where $\mathbf{T}$ is the unit tangent vector to the space curve, $\mathbf{N}$ is its unit principal normal, and ρ is the radius of curvature.

Velocity $\mathbf{v}$ = magnitude of $\mathbf{v}$ multiplied by unit tangent vector $\mathbf{T}$

or $\mathbf{v} = v\mathbf{T}$

Differentiating, $$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d}{dt}(v\mathbf{T}) = \frac{dv}{dt}\mathbf{T} + v\frac{d\mathbf{T}}{dt}$$

But by Problem 18(*a*), $$\frac{d\mathbf{T}}{dt} = \frac{d\mathbf{T}}{ds}\frac{ds}{dt} = \kappa\mathbf{N}\frac{ds}{dt} = \kappa v\mathbf{N} = \frac{v\mathbf{N}}{\rho}$$

Then $$\mathbf{a} = \frac{dv}{dt}\mathbf{T} + v\left(\frac{v\mathbf{N}}{\rho}\right) = \frac{dv}{dt}\mathbf{T} + \frac{v^2}{\rho}\mathbf{N}$$

This shows that the component of the acceleration is dv/dt in a direction tangent to the path and v^2/ρ in a direction of the principal normal to the path. The latter acceleration is often called the *centripetal acceleration*. For a special case of this problem see Problem 12.

28. If $\mathbf{r}$ is the position vector of a particle of mass m relative to point O and $\mathbf{F}$ is the external force on the particle, then $\mathbf{r}\times\mathbf{F} = \mathbf{M}$ is the torque or moment of $\mathbf{F}$ about O. Show that $\mathbf{M} = d\mathbf{H}/dt$, where $\mathbf{H} = \mathbf{r}\times m\mathbf{v}$ and $\mathbf{v}$ is the velocity of the particle.

$$\mathbf{M} = \mathbf{r}\times\mathbf{F} = \mathbf{r}\times\frac{d}{dt}(m\mathbf{v}) \qquad \text{by Newton's law.}$$

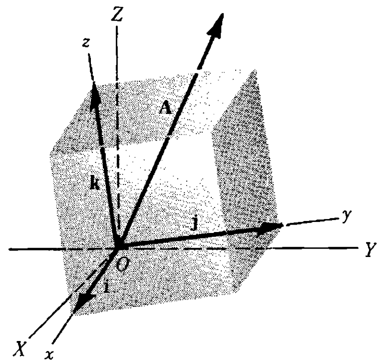
But $$\frac{d}{dt}(\mathbf{r}\times m\mathbf{v}) = \mathbf{r}\times\frac{d}{dt}(m\mathbf{v}) + \frac{d\mathbf{r}}{dt}\times m\mathbf{v}$$
$$= \mathbf{r}\times\frac{d}{dt}(m\mathbf{v}) + \mathbf{v}\times m\mathbf{v} = \mathbf{r}\times\frac{d}{dt}(m\mathbf{v}) + \mathbf{0}$$

i.e. $$\mathbf{M} = \frac{d}{dt}(\mathbf{r}\times m\mathbf{v}) = \frac{d\mathbf{H}}{dt}$$

Note that the result holds whether m is constant or not. $\mathbf{H}$ is called the *angular momentum*. The result states that the torque is equal to the time rate of change of angular momentum.

This result is easily extended to a system of n particles having respective masses $m_1, m_2, \ldots, m_n$ and position vectors $\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_n$ with external forces $\mathbf{F}_1, \mathbf{F}_2, \ldots, \mathbf{F}_n$. For this case, $\mathbf{H} = \sum_{k=1}^{n} m_k \mathbf{r}_k \times \mathbf{v}_k$ is the total angular momentum, $\mathbf{M} = \sum_{k=1}^{n} \mathbf{r}_k \times \mathbf{F}_k$ is the total torque, and the result is $\mathbf{M} = \dfrac{d\mathbf{H}}{dt}$ as before.

29. An observer stationed at a point which is fixed relative to an xyz coordinate system with origin O, as shown in the adjoining diagram, observes a vector $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$ and calculates its time derivative to be $\dfrac{dA_1}{dt}\mathbf{i} + \dfrac{dA_2}{dt}\mathbf{j} + \dfrac{dA_3}{dt}\mathbf{k}$. Later, he finds out that he and his coordinate system are actually rotating with respect to an XYZ coordinate system taken as fixed in space and having origin also at O. He asks, 'What would be the time derivative of $\mathbf{A}$ for an observer who is fixed relative to the XYZ coordinate system?'



(a) If $\left.\dfrac{d\mathbf{A}}{dt}\right|_f$ and $\left.\dfrac{d\mathbf{A}}{dt}\right|_m$ denote respectively the time derivatives of $\mathbf{A}$ with respect to the fixed and moving systems, show that there exists a vector quantity $\boldsymbol{\omega}$ such that

$$\left.\frac{d\mathbf{A}}{dt}\right|_f = \left.\frac{d\mathbf{A}}{dt}\right|_m + \boldsymbol{\omega} \times \mathbf{A}$$

(b) Let D_f and D_m be symbolic time derivative operators in the fixed and moving systems respectively. Demonstrate the operator equivalence

$$D_f \equiv D_m + \boldsymbol{\omega} \times$$

(a) To the fixed observer the unit vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$ actually change with time. Hence such an observer would compute the time derivative of $\mathbf{A}$ as

$$(1) \qquad \frac{d\mathbf{A}}{dt} = \frac{dA_1}{dt}\mathbf{i} + \frac{dA_2}{dt}\mathbf{j} + \frac{dA_3}{dt}\mathbf{k} + A_1\frac{d\mathbf{i}}{dt} + A_2\frac{d\mathbf{j}}{dt} + A_3\frac{d\mathbf{k}}{dt} \qquad \text{i.e.}$$

$$(2) \qquad \left.\frac{d\mathbf{A}}{dt}\right|_f = \left.\frac{d\mathbf{A}}{dt}\right|_m + A_1\frac{d\mathbf{i}}{dt} + A_2\frac{d\mathbf{j}}{dt} + A_3\frac{d\mathbf{k}}{dt}$$

Since $\mathbf{i}$ is a unit vector, $d\mathbf{i}/dt$ is perpendicular to $\mathbf{i}$ (see Problem 9) and must therefore lie in the plane of $\mathbf{j}$ and $\mathbf{k}$. Then

$$(3) \qquad \frac{d\mathbf{i}}{dt} = \alpha_1\mathbf{j} + \alpha_2\mathbf{k}$$

Similarly,

$$(4) \qquad \frac{d\mathbf{j}}{dt} = \alpha_3\mathbf{k} + \alpha_4\mathbf{i}$$

$$(5) \qquad \frac{d\mathbf{k}}{dt} = \alpha_5\mathbf{i} + \alpha_6\mathbf{j}$$

From $\mathbf{i}\cdot\mathbf{j} = 0$, differentiation yields $\mathbf{i}\cdot\dfrac{d\mathbf{j}}{dt} + \dfrac{d\mathbf{i}}{dt}\cdot\mathbf{j} = 0$. But $\mathbf{i}\cdot\dfrac{d\mathbf{j}}{dt} = \alpha_4$ from (4), and $\dfrac{d\mathbf{i}}{dt}\cdot\mathbf{j} = \alpha_1$ from (3); then $\alpha_4 = -\alpha_1$.

Similarly from $\mathbf{i}\cdot\mathbf{k} = 0$, $\mathbf{i}\cdot\dfrac{d\mathbf{k}}{dt} + \dfrac{d\mathbf{i}}{dt}\cdot\mathbf{k} = 0$ and $\alpha_5 = -\alpha_2$; from $\mathbf{j}\cdot\mathbf{k} = 0$, $\mathbf{j}\cdot\dfrac{d\mathbf{k}}{dt} + \dfrac{d\mathbf{j}}{dt}\cdot\mathbf{k} = 0$ and $\alpha_6 = -\alpha_3$.

Then $\dfrac{d\mathbf{i}}{dt} = \alpha_1\mathbf{j} + \alpha_2\mathbf{k}$, $\dfrac{d\mathbf{j}}{dt} = \alpha_3\mathbf{k} - \alpha_1\mathbf{i}$, $\dfrac{d\mathbf{k}}{dt} = -\alpha_2\mathbf{i} - \alpha_3\mathbf{j}$ and

$$A_1\frac{d\mathbf{i}}{dt} + A_2\frac{d\mathbf{j}}{dt} + A_3\frac{d\mathbf{k}}{dt} = (-\alpha_1 A_2 - \alpha_2 A_3)\mathbf{i} + (\alpha_1 A_1 - \alpha_3 A_3)\mathbf{j} + (\alpha_2 A_1 + \alpha_3 A_2)\mathbf{k}$$

which can be written as

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \alpha_3 & -\alpha_2 & \alpha_1 \\ A_1 & A_2 & A_3 \end{vmatrix}$$

Then if we choose $\alpha_3 = \omega_1$, $-\alpha_2 = \omega_2$, $\alpha_1 = \omega_3$ the determinant becomes

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \omega_1 & \omega_2 & \omega_3 \\ A_1 & A_2 & A_3 \end{vmatrix} = \boldsymbol{\omega} \times \mathbf{A}$$

where $\boldsymbol{\omega} = \omega_1\mathbf{i} + \omega_2\mathbf{j} + \omega_3\mathbf{k}$. The quantity $\boldsymbol{\omega}$ is the angular velocity vector of the moving system with respect to the fixed system.

(*b*) By definition $D_f\mathbf{A} = \left.\frac{d\mathbf{A}}{dt}\right|_f$ = derivative in fixed system

$D_m\mathbf{A} = \left.\frac{d\mathbf{A}}{dt}\right|_m$ = derivative in moving system.

From (*a*), $$D_f\mathbf{A} = D_m\mathbf{A} + \boldsymbol{\omega}\times\mathbf{A} = (D_m + \boldsymbol{\omega}\times)\mathbf{A}$$

and shows the equivalence of the operators $D_f \equiv D_m + \boldsymbol{\omega}\times$.

30. Determine the (*a*) velocity and (*b*) acceleration of a moving particle as seen by the two observers in Problem 29.

(*a*) Let vector $\mathbf{A}$ in Problem 29 be the position vector $\mathbf{r}$ of the particle. Using the operator notation of Problem 29(*b*), we have

(*1*) $$D_f\mathbf{r} = (D_m + \boldsymbol{\omega}\times)\mathbf{r} = D_m\mathbf{r} + \boldsymbol{\omega}\times\mathbf{r}$$

But $D_f\mathbf{r} = \mathbf{v}_{p|f}$ = velocity of particle relative to fixed system

$D_m\mathbf{r} = \mathbf{v}_{p|m}$ = velocity of particle relative to moving system

$\boldsymbol{\omega}\times\mathbf{r} = \mathbf{v}_{m|f}$ = velocity of moving system relative to fixed system.

Then (*1*) can be written as

(*2*) $$\mathbf{v}_{p|f} = \mathbf{v}_{p|m} + \boldsymbol{\omega}\times\mathbf{r}$$

or in the suggestive notation

(*3*) $$\mathbf{v}_{p|f} = \mathbf{v}_{p|m} + \mathbf{v}_{m|f}$$

Note that the roles of fixed and moving observers can, of course, be interchanged. Thus the fixed observer can think of himself as really moving with respect to the other. For this case we must interchange subscripts m and f and also change $\boldsymbol{\omega}$ to $-\boldsymbol{\omega}$ since the relative rotation is reversed. If this is done, (*2*) becomes

$$\mathbf{v}_{p|m} = \mathbf{v}_{p|f} - \boldsymbol{\omega}\times\mathbf{r} \quad \text{or} \quad \mathbf{v}_{p|f} = \mathbf{v}_{p|m} + \boldsymbol{\omega}\times\mathbf{r}$$

so that the result is valid for each observer.

(*b*) The acceleration of the particle as determined by the fixed observer at O is $D_f^2\mathbf{r} = D_f(D_f\mathbf{r})$. Take D_f of both sides of (*1*), using the operator equivalence established in Problem 29(*b*). Then

$$\begin{aligned} D_f(D_f\mathbf{r}) &= D_f(D_m\mathbf{r} + \boldsymbol{\omega}\times\mathbf{r}) \\ &= (D_m + \boldsymbol{\omega}\times)(D_m\mathbf{r} + \boldsymbol{\omega}\times\mathbf{r}) \\ &= D_m(D_m\mathbf{r} + \boldsymbol{\omega}\times\mathbf{r}) + \boldsymbol{\omega}\times(D_m\mathbf{r} + \boldsymbol{\omega}\times\mathbf{r}) \\ &= D_m^2\mathbf{r} + D_m(\boldsymbol{\omega}\times\mathbf{r}) + \boldsymbol{\omega}\times D_m\mathbf{r} + \boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{r}) \end{aligned}$$

or

$$D_f^2\mathbf{r} = D_m^2\mathbf{r} + 2\boldsymbol{\omega}\times D_m\mathbf{r} + (D_m\boldsymbol{\omega})\times\mathbf{r} + \boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{r})$$

Let $\mathbf{a}_{p|f} = D_f^2\mathbf{r}$ = acceleration of particle relative to fixed system

$\mathbf{a}_{p|m} = D_m^2\mathbf{r}$ = acceleration of particle relative to moving system.

Then $\mathbf{a}_{m|f} = 2\boldsymbol{\omega}\times D_m\mathbf{r} + (D_m\boldsymbol{\omega})\times\mathbf{r} + \boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{r})$

= acceleration of moving system relative to fixed system

and we can write

$$\mathbf{a}_{p|f} = \mathbf{a}_{p|m} + \mathbf{a}_{m|f}\,.$$

For many cases of importance $\boldsymbol{\omega}$ is a constant vector, i.e. the rotation proceeds with constant angular velocity. Then $D_m\boldsymbol{\omega} = \mathbf{0}$ and

$$\mathbf{a}_{m|f} = 2\boldsymbol{\omega}\times D_m\mathbf{r} + \boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{r}) = 2\boldsymbol{\omega}\times\mathbf{v}_m + \boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{r})$$

The quantity $2\boldsymbol{\omega}\times\mathbf{v}_m$ is called the *Coriolis acceleration* and $\boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{r})$ is called the *centripetal acceleration*.

Newton's laws are strictly valid only in *inertial systems*, i.e. systems which are either fixed or which move with constant velocity relative to a fixed system. The earth is not exactly an inertial system and this accounts for the presence of the so called 'fictitious' extra forces (Coriolis, etc.) which must be considered. If the mass of a particle is a constant M, then Newton's second law becomes

$$(4)\qquad M D_m^2\mathbf{r} = \mathbf{F} - 2M(\boldsymbol{\omega}\times D_m\mathbf{r}) - M[\boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{r})]$$

where D_m denotes d/dt as computed by an observer on the earth, and $\mathbf{F}$ is the resultant of all real forces as measured by this observer. The last two terms on the right of (*4*) are negligible in most cases and are not used in practice.

The *theory of relativity* due to Einstein has modified quite radically the concepts of absolute motion which are implied by Newtonian concepts and has led to revision of Newton's laws.

SUPPLEMENTARY PROBLEMS

31. If $\mathbf{R} = e^{-t}\mathbf{i} + \ln(t^2+1)\mathbf{j} - \tan t\,\mathbf{k}$, find (*a*) $\dfrac{d\mathbf{R}}{dt}$, (*b*) $\dfrac{d^2\mathbf{R}}{dt^2}$, (*c*) $\left|\dfrac{d\mathbf{R}}{dt}\right|$, (*d*) $\left|\dfrac{d^2\mathbf{R}}{dt^2}\right|$ at $t=0$.
Ans. (*a*) $-\mathbf{i}-\mathbf{k}$, (*b*) $\mathbf{i}+2\mathbf{j}$, (*c*) $\sqrt{2}$, (*d*) $\sqrt{5}$

32. Find the velocity and acceleration of a particle which moves along the curve $x = 2\sin 3t$, $y = 2\cos 3t$, $z = 8t$ at any time $t > 0$. Find the magnitude of the velocity and acceleration.
Ans. $\mathbf{v} = 6\cos 3t\,\mathbf{i} - 6\sin 3t\,\mathbf{j} + 8\mathbf{k}$, $\mathbf{a} = -18\sin 3t\,\mathbf{i} - 18\cos 3t\,\mathbf{j}$, $|\mathbf{v}| = 10$, $|\mathbf{a}| = 18$

33. Find a unit tangent vector to any point on the curve $x = a\cos\omega t$, $y = a\sin\omega t$, $z = bt$ where a, b, ω are constants. *Ans.* $\dfrac{-a\omega\sin\omega t\,\mathbf{i} + a\omega\cos\omega t\,\mathbf{j} + b\mathbf{k}}{\sqrt{a^2\omega^2+b^2}}$

34. If $\mathbf{A} = t^2\mathbf{i} - t\mathbf{j} + (2t+1)\mathbf{k}$ and $\mathbf{B} = (2t-3)\mathbf{i} + \mathbf{j} - t\mathbf{k}$, find
(*a*) $\dfrac{d}{dt}(\mathbf{A}\cdot\mathbf{B})$, (*b*) $\dfrac{d}{dt}(\mathbf{A}\times\mathbf{B})$, (*c*) $\dfrac{d}{dt}|\mathbf{A}+\mathbf{B}|$, (*d*) $\dfrac{d}{dt}(\mathbf{A}\times\dfrac{d\mathbf{B}}{dt})$ at $t=1$. *Ans.* (*a*) -6, (*b*) $7\mathbf{j}+3\mathbf{k}$, (*c*) 1, (*d*) $\mathbf{i}+6\mathbf{j}+2\mathbf{k}$

35. If $\mathbf{A} = \sin u\,\mathbf{i} + \cos u\,\mathbf{j} + u\,\mathbf{k}$, $\mathbf{B} = \cos u\,\mathbf{i} - \sin u\,\mathbf{j} - 3\mathbf{k}$, and $\mathbf{C} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$, find $\frac{d}{du}(\mathbf{A}\times(\mathbf{B}\times\mathbf{C}))$ at $u=0$.
Ans. $7\mathbf{i} + 6\mathbf{j} - 6\mathbf{k}$

36. Find $\frac{d}{ds}(\mathbf{A}\cdot\frac{d\mathbf{B}}{ds} - \frac{d\mathbf{A}}{ds}\cdot\mathbf{B})$ if $\mathbf{A}$ and $\mathbf{B}$ are differentiable functions of s. *Ans.* $\mathbf{A}\cdot\frac{d^2\mathbf{B}}{ds^2} - \frac{d^2\mathbf{A}}{ds^2}\cdot\mathbf{B}$

37. If $\mathbf{A}(t) = 3t^2\,\mathbf{i} - (t+4)\mathbf{j} + (t^2-2t)\mathbf{k}$ and $\mathbf{B}(t) = \sin t\,\mathbf{i} + 3e^{-t}\mathbf{j} - 3\cos t\,\mathbf{k}$, find $\frac{d^2}{dt^2}(\mathbf{A}\times\mathbf{B})$ at $t=0$.
Ans. $-30\mathbf{i} + 14\mathbf{j} + 20\mathbf{k}$

38. If $\frac{d^2\mathbf{A}}{dt^2} = 6t\,\mathbf{i} - 24t^2\,\mathbf{j} + 4\sin t\,\mathbf{k}$, find $\mathbf{A}$ given that $\mathbf{A} = 2\mathbf{i} + \mathbf{j}$ and $\frac{d\mathbf{A}}{dt} = -\mathbf{i} - 3\mathbf{k}$ at $t=0$.
Ans. $\mathbf{A} = (t^3 - t + 2)\mathbf{i} + (1 - 2t^4)\mathbf{j} + (t - 4\sin t)\mathbf{k}$

39. Show that $\mathbf{r} = e^{-t}(\mathbf{C}_1\cos 2t + \mathbf{C}_2\sin 2t)$, where $\mathbf{C}_1$ and $\mathbf{C}_2$ are constant vectors, is a solution of the differential equation $\frac{d^2\mathbf{r}}{dt^2} + 2\frac{d\mathbf{r}}{dt} + 5\mathbf{r} = \mathbf{0}$.

40. Show that the general solution of the differential equation $\frac{d^2\mathbf{r}}{dt^2} + 2\alpha\frac{d\mathbf{r}}{dt} + \omega^2\mathbf{r} = \mathbf{0}$, where α and ω are constants, is

(*a*) $\mathbf{r} = e^{-\alpha t}(\mathbf{C}_1 e^{\sqrt{\alpha^2-\omega^2}\,t} + \mathbf{C}_2 e^{-\sqrt{\alpha^2-\omega^2}\,t})$ if $\alpha^2 - \omega^2 > 0$

(*b*) $\mathbf{r} = e^{-\alpha t}(\mathbf{C}_1\sin\sqrt{\omega^2-\alpha^2}\,t + \mathbf{C}_2\cos\sqrt{\omega^2-\alpha^2}\,t)$ if $\alpha^2 - \omega^2 < 0$.

(*c*) $\mathbf{r} = e^{-\alpha t}(\mathbf{C}_1 + \mathbf{C}_2 t)$ if $\alpha^2 - \omega^2 = 0$,

where $\mathbf{C}_1$ and $\mathbf{C}_2$ are arbitrary constant vectors.

41. Solve (*a*) $\frac{d^2\mathbf{r}}{dt^2} - 4\frac{d\mathbf{r}}{dt} - 5\mathbf{r} = \mathbf{0}$, (*b*) $\frac{d^2\mathbf{r}}{dt^2} + 2\frac{d\mathbf{r}}{dt} + \mathbf{r} = \mathbf{0}$, (*c*) $\frac{d^2\mathbf{r}}{dt^2} + 4\mathbf{r} = \mathbf{0}$.
Ans. (*a*) $\mathbf{r} = \mathbf{C}_1 e^{5t} + \mathbf{C}_2 e^{-t}$, (*b*) $\mathbf{r} = e^{-t}(\mathbf{C}_1 + \mathbf{C}_2 t)$, (*c*) $\mathbf{r} = \mathbf{C}_1\cos 2t + \mathbf{C}_2\sin 2t$

42. Solve $\frac{d\mathbf{Y}}{dt} = \mathbf{X}$, $\frac{d\mathbf{X}}{dt} = -\mathbf{Y}$. *Ans.* $\mathbf{X} = \mathbf{C}_1\cos t + \mathbf{C}_2\sin t$, $\mathbf{Y} = \mathbf{C}_1\sin t - \mathbf{C}_2\cos t$

43. If $\mathbf{A} = \cos xy\,\mathbf{i} + (3xy - 2x^2)\mathbf{j} - (3x + 2y)\mathbf{k}$, find $\frac{\partial\mathbf{A}}{\partial x}, \frac{\partial\mathbf{A}}{\partial y}, \frac{\partial^2\mathbf{A}}{\partial x^2}, \frac{\partial^2\mathbf{A}}{\partial y^2}, \frac{\partial^2\mathbf{A}}{\partial x\,\partial y}, \frac{\partial^2\mathbf{A}}{\partial y\,\partial x}$.

Ans. $\frac{\partial\mathbf{A}}{\partial x} = -y\sin xy\,\mathbf{i} + (3y - 4x)\mathbf{j} - 3\mathbf{k}$, $\frac{\partial\mathbf{A}}{\partial y} = -x\sin xy\,\mathbf{i} + 3x\,\mathbf{j} - 2\mathbf{k}$,

$\frac{\partial^2\mathbf{A}}{\partial x^2} = -y^2\cos xy\,\mathbf{i} - 4\mathbf{j}$, $\frac{\partial^2\mathbf{A}}{\partial y^2} = -x^2\cos xy\,\mathbf{i}$, $\frac{\partial^2\mathbf{A}}{\partial x\,\partial y} = \frac{\partial^2\mathbf{A}}{\partial y\,\partial x} = -(xy\cos xy + \sin xy)\mathbf{i} + 3\mathbf{j}$

44. If $\mathbf{A} = x^2yz\,\mathbf{i} - 2xz^3\,\mathbf{j} + xz^2\,\mathbf{k}$ and $\mathbf{B} = 2z\,\mathbf{i} + y\,\mathbf{j} - x^2\,\mathbf{k}$, find $\frac{\partial^2}{\partial x\,\partial y}(\mathbf{A}\times\mathbf{B})$ at $(1,0,-2)$.
Ans. $-4\mathbf{i} - 8\mathbf{j}$

45. If $\mathbf{C}_1$ and $\mathbf{C}_2$ are constant vectors and λ is a constant scalar, show that $\mathbf{H} = e^{-\lambda x}(\mathbf{C}_1\sin\lambda y + \mathbf{C}_2\cos\lambda y)$ satisfies the partial differential equation $\frac{\partial^2\mathbf{H}}{\partial x^2} + \frac{\partial^2\mathbf{H}}{\partial y^2} = \mathbf{0}$.

46. Prove that $\mathbf{A} = \frac{\mathbf{p}_0\, e^{i\omega(t - r/c)}}{r}$, where $\mathbf{p}_0$ is a constant vector, ω and c are constant scalars and $i = \sqrt{-1}$, satisfies the equation $\frac{\partial^2\mathbf{A}}{\partial r^2} + \frac{2}{r}\frac{\partial\mathbf{A}}{\partial r} = \frac{1}{c^2}\frac{\partial^2\mathbf{A}}{\partial t^2}$. This result is of importance in *electromagnetic theory*.

DIFFERENTIAL GEOMETRY

47. Find (*a*) the unit tangent $\mathbf{T}$, (*b*) the curvature κ, (*c*) the principal normal $\mathbf{N}$, (*d*) the binormal $\mathbf{B}$, and (*e*) the torsion τ for the space curve $x = t - t^3/3$, $y = t^2$, $z = t + t^3/3$.

Ans. (*a*) $\mathbf{T} = \frac{(1-t^2)\mathbf{i} + 2t\,\mathbf{j} + (1+t^2)\mathbf{k}}{\sqrt{2}(1+t^2)}$ (*c*) $\mathbf{N} = -\frac{2t}{1+t^2}\mathbf{i} + \frac{1-t^2}{1+t^2}\mathbf{j}$ (*e*) $\tau = \frac{1}{(1+t^2)^2}$

(*b*) $\kappa = \frac{1}{(1+t^2)^2}$ (*d*) $\mathbf{B} = \frac{(t^2-1)\mathbf{i} - 2t\,\mathbf{j} + (t^2+1)\mathbf{k}}{\sqrt{2}(1+t^2)}$

48. A space curve is defined in terms of the arc length parameter s by the equations

$$x = \arc\tan s, \quad y = \tfrac{1}{2}\sqrt{2}\ln(s^2+1), \quad z = s - \arc\tan s$$

Find (*a*) $\mathbf{T}$, (*b*) $\mathbf{N}$, (*c*) $\mathbf{B}$, (*d*) κ, (*e*) τ, (*f*) ρ, (*g*) σ.

Ans. (*a*) $\mathbf{T} = \dfrac{\mathbf{i} + \sqrt{2}s\,\mathbf{j} + s^2\mathbf{k}}{s^2+1}$ (*d*) $\kappa = \dfrac{\sqrt{2}}{s^2+1}$

(*b*) $\mathbf{N} = \dfrac{-\sqrt{2}s\,\mathbf{i} + (1-s^2)\mathbf{j} + \sqrt{2}s\,\mathbf{k}}{s^2+1}$ (*e*) $\tau = \dfrac{\sqrt{2}}{s^2+1}$ (*g*) $\sigma = \dfrac{s^2+1}{\sqrt{2}}$

(*c*) $\mathbf{B} = \dfrac{s^2\mathbf{i} - \sqrt{2}s\,\mathbf{j} + \mathbf{k}}{s^2+1}$ (*f*) $\rho = \dfrac{s^2+1}{\sqrt{2}}$

49. Find κ and τ for the space curve $x = t,\ y = t^2,\ z = t^3$ called the twisted cubic.

Ans. $\kappa = \dfrac{2\sqrt{9t^4 + 9t^2 + 1}}{(9t^4 + 4t^2 + 1)^{3/2}}, \quad \tau = \dfrac{3}{9t^4 + 9t^2 + 1}$

50. Show that for a plane curve the torsion $\tau = 0$.

51. Show that the radius of curvature of a plane curve with equations $y = f(x)$, $z = 0$, i.e. a curve in the xy plane is given by $\rho = \dfrac{[1 + (y')^2]^{3/2}}{|y''|}$.

52. Find the curvature and radius of curvature of the curve with position vector $\mathbf{r} = a\cos u\,\mathbf{i} + b\sin u\,\mathbf{j}$, where a and b are positive constants. Interpret the case where $a = b$.

Ans. $\kappa = \dfrac{ab}{(a^2\sin^2 u + b^2\cos^2 u)^{3/2}} = \dfrac{1}{\rho}$; if $a = b$, the given curve which is an ellipse, becomes a circle of radius a and its radius of curvature $\rho = a$.

53. Show that the Frenet-Serret formulas can be written in the form $\dfrac{d\mathbf{T}}{ds} = \boldsymbol{\omega}\times\mathbf{T}$, $\dfrac{d\mathbf{N}}{ds} = \boldsymbol{\omega}\times\mathbf{N}$, $\dfrac{d\mathbf{B}}{ds} = \boldsymbol{\omega}\times\mathbf{B}$ and determine $\boldsymbol{\omega}$. *Ans.* $\boldsymbol{\omega} = \tau\mathbf{T} + \kappa\mathbf{B}$

54. Prove that the curvature of the space curve $\mathbf{r} = \mathbf{r}(t)$ is given numerically by $\kappa = \dfrac{|\dot{\mathbf{r}}\times\ddot{\mathbf{r}}|}{|\dot{\mathbf{r}}|^3}$, where dots denote differentiation with respect to t.

55. (*a*) Prove that $\tau = \dfrac{\dot{\mathbf{r}}\cdot\ddot{\mathbf{r}}\times\dddot{\mathbf{r}}}{|\dot{\mathbf{r}}\times\ddot{\mathbf{r}}|^2}$ for the space curve $\mathbf{r} = \mathbf{r}(t)$.

(*b*) If the parameter t is the arc length s show that $\tau = \dfrac{\dfrac{d\mathbf{r}}{ds}\cdot\dfrac{d^2\mathbf{r}}{ds^2}\times\dfrac{d^3\mathbf{r}}{ds^3}}{(d^2\mathbf{r}/ds^2)^2}$.

56. If $\mathbf{Q} = \dot{\mathbf{r}}\times\ddot{\mathbf{r}}$, show that $\kappa = \dfrac{Q}{|\dot{\mathbf{r}}|^3}$, $\tau = \dfrac{\mathbf{Q}\cdot\dddot{\mathbf{r}}}{Q^2}$.

57. Find κ and τ for the space curve $x = \theta - \sin\theta$, $y = 1 - \cos\theta$, $z = 4\sin(\theta/2)$.

Ans. $\kappa = \dfrac{1}{8}\sqrt{6 - 2\cos\theta}, \quad \tau = \dfrac{(3+\cos\theta)\cos\theta/2 + 2\sin\theta\sin\theta/2}{12\cos\theta - 4}$

58. Find the torsion of the curve $x = \dfrac{2t+1}{t-1}$, $y = \dfrac{t^2}{t-1}$, $z = t + 2$. Explain your answer.

Ans. $\tau = 0$. The curve lies on the plane $x - 3y + 3z = 5$.

59. Show that the equations of the tangent line, principal normal and binormal to the space curve $\mathbf{r} = \mathbf{r}(t)$ at the point $t = t_0$ can be written respectively $\mathbf{r} = \mathbf{r}_0 + t\mathbf{T}_0$, $\mathbf{r} = \mathbf{r}_0 + t\mathbf{N}_0$, $\mathbf{r} = \mathbf{r}_0 + t\mathbf{B}_0$, where t is a parameter.

60. Find equations for the (*a*) tangent, (*b*) principal normal and (*c*) binormal to the curve $x = 3\cos t$, $y = 3\sin t$, $z = 4t$ at the point where $t = \pi$.

Ans. (*a*) Tangent: $\mathbf{r} = -3\mathbf{i} + 4\pi\mathbf{k} + t(-\frac{3}{5}\mathbf{j} + \frac{4}{5}\mathbf{k})$ or $x = -3,\ y = -\frac{3}{5}t,\ z = 4\pi + \frac{4}{5}t$.

(b) Normal: $\mathbf{r} = -3\mathbf{i} + 4\pi\mathbf{j} + t\,\mathbf{i}$ or $x = -3 + t,\ y = 4\pi,\ z = 0$.

(c) Binormal: $\mathbf{r} = -3\mathbf{i} + 4\pi\mathbf{j} + t(\frac{4}{5}\mathbf{j} + \frac{3}{5}\mathbf{k})$ or $x = -3,\ y = 4\pi + \frac{4}{5}t,\ z = \frac{3}{5}t$.

61. Find equations for the (a) osculating plane, (b) normal plane and (c) rectifying plane to the curve $x = 3t - t^3$, $y = 3t^2$, $z = 3t + t^3$ at the point where $t = 1$. *Ans.* (a) $y - z + 1 = 0$, (b) $y + z - 7 = 0$, (c) $x = 2$

62. (a) Show that the differential of arc length on the surface $\mathbf{r} = \mathbf{r}(u,v)$ is given by

$$ds^2 = E\,du^2 + 2F\,du\,dv + G\,dv^2$$

where $E = \frac{\partial \mathbf{r}}{\partial u}\cdot\frac{\partial \mathbf{r}}{\partial u} = (\frac{\partial \mathbf{r}}{\partial u})^2$, $F = \frac{\partial \mathbf{r}}{\partial u}\cdot\frac{\partial \mathbf{r}}{\partial v}$, $G = \frac{\partial \mathbf{r}}{\partial v}\cdot\frac{\partial \mathbf{r}}{\partial v} = (\frac{\partial \mathbf{r}}{\partial v})^2$.

(b) Prove that a necessary and sufficient condition that the u,v curvilinear coordinate system be orthogonal is $F \equiv 0$.

63. Find an equation for the tangent plane to the surface $z = xy$ at the point (2,3,6). *Ans.* $3x + 2y - z = 6$

64. Find equations for the tangent plane and normal line to the surface $4z = x^2 - y^2$ at the point (3,1,2).
Ans. $3x - y - 2z = 4$; $x = 3t + 3,\ y = 1 - t,\ z = 2 - 2t$

65. Prove that a unit normal to the surface $\mathbf{r} = \mathbf{r}(u,v)$ is $\mathbf{n} = \pm \dfrac{\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v}}{\sqrt{EG - F^2}}$, where E, F, and G are defined as in Problem 62.

MECHANICS

66. A particle moves along the curve $\mathbf{r} = (t^3 - 4t)\mathbf{i} + (t^2 + 4t)\mathbf{j} + (8t^2 - 3t^3)\mathbf{k}$, where t is the time. Find the magnitudes of the tangential and normal components of its acceleration when $t = 2$.
Ans. Tangential, 16 ; normal, $2\sqrt{73}$

67. If a particle has velocity $\mathbf{v}$ and acceleration $\mathbf{a}$ along a space curve, prove that the radius of curvature of its path is given numerically by $\rho = \dfrac{v^3}{|\mathbf{v} \times \mathbf{a}|}$.

68. An object is attracted to a fixed point O with a force $\mathbf{F} = f(r)\mathbf{r}$, called a *central force*, where $\mathbf{r}$ is the position vector of the object relative to O. Show that $\mathbf{r} \times \mathbf{v} = \mathbf{h}$ where $\mathbf{h}$ is a constant vector. Prove that the angular momentum is constant.

69. Prove that the acceleration vector of a particle moving along a space curve always lies in the osculating plane.

70. (a) Find the acceleration of a particle moving in the xy plane in terms of polar coordinates (ρ, ϕ).
(b) What are the components of the acceleration parallel and perpendicular to ρ ?

Ans. (a) $\ddot{\mathbf{r}} = [(\ddot{\rho} - \rho\dot{\phi}^2)\cos\phi - (\rho\ddot{\phi} + 2\dot{\rho}\dot{\phi})\sin\phi]\,\mathbf{i} + [(\ddot{\rho} - \rho\dot{\phi}^2)\sin\phi + (\rho\ddot{\phi} + 2\dot{\rho}\dot{\phi})\cos\phi]\,\mathbf{j}$

(b) $\ddot{\rho} - \rho\dot{\phi}^2$, $\rho\ddot{\phi} + 2\dot{\rho}\dot{\phi}$